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Low-dimensional Control under Embodiment Constraints: From LQR to Reinforcement Learning

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Abstract

Your English abstract here (mandatory if written in English and recommended otherwise).

Zusammenfassung

Hier die deutsche Zusammenfassung einfügen (notwendig).

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Chapter 1

Introduction

Over the recent years, the field of robotics has seen rapid advancements in the capabilities of neural-network-based implementations. An increase in global compute power[CITE: TODO compute], fueled by the so-called "AI race" of LLMs, has allowed researchers around the world to create even more extensive model architectures for more complex use cases [CITE: TODO llm example]. Transferring these structuresto physical bodies requires more space and on-device compute, often the limiting factors of embodiment. To bridge this gap between powerful models and limited resources, the approach of cheap design has become more common. As mentioned by N. Ay[CITE: TODO ay geometric], the core idea is to maintain an "embodied universal approximator", while reducing the overall complexity of the model, which can be defined based on the specific case. The most straightforward approach is inducing the underlying physical dynamics of the embodiment into the architecture, which is often of lower dimensionality than the actual sensors and actuators. Recent papers focusing on algorithms for the LQR[CITE: TODO lqr], apply this approach by offering implementations for enforcing low-rank controllers, leading to lower complexity in relation to parameterization. Still, this low-dimensional approach has yet to be extended[CITE: TODO extension] to more complex embodiments, and is missing a crucial part required for implementation: Robustness. This leaves us with the question: *Is it possible to train a low-dimensional controller for an embodied agent, that is performant yet robust?*

As robustness is a wide topic, in our case, we use it when referring to sensor- and actuator noise, as well as parametric uncertainties of the underlying model and external disturbances. When talking about performance, we mean reducing cost as much as possible in relation to an optimal controller. In the end, we want to show in which cases low-dimensional controllers are a viable alternative, and can thus reduce costs for hardware implementations.

To answer our question, we will first explain the core concepts and collect existing approaches for the LQR setting in Chapter 2. In Chapter 3, we will compare these implementations using the double integrator problem for LQR with respect to gen-

eral performance, as well as robustness. Building on that, we will artificially inflate our inputs and outputs via matrix multiplication in Chapter 4 and additionally introduce deep RL baseline models into the comparison. For Chapter 5, we will completely move away from LQR, and extend the setting to a perception-based implementation. Here we will compare different models with varying parameter complexity in regard to robustness, applying the concepts we learned from our earlier experiments. Finally, we will conclude by analyzing whether we indeed achieved our goal of constructing a robust controller, and stating how our approach can be applied to more complex settings.

Chapter 2

Background

2.1 Preliminaries

Before turning to the experiments, this chapter revisits the concepts that all later parts of this thesis build on. Section 2.1.1 introduces linear state-space models and their discretization, Section 2.1.2 shows them on the double integrator that serves as our running example throughout, and Sections 2.1.3 and 2.1.4 recall the stability notions and the linear quadratic regulator. Sections 2.3.1.2 and 2.3.2 define the perturbation models and the robustness metrics used to evaluate every controller we construct.

2.1.1 Linear State-Space Models

In control theory, the evolution of an agent together with its environment is described by a *state-space model*: a rule that assigns to every point of a state space the instantaneous rate at which the system moves through it. Throughout this thesis we restrict attention to the class of *linear time-invariant* (LTI) systems, for which this rule is linear in both the state and the input and does not itself depend on time[CITE: ControlSystemsTheoryAndDesign.pdf].

A continuous-time LTI system is given by

$$\dot{x}(t) = A_c\, x(t) + B_c\, u(t), \qquad x(0) = x_0, \tag{2.1a}$$

$$y(t) = C\, x(t) + D\, u(t), \tag{2.1b}$$

where $x(t) \in \mathbb{R}^n$ is the *state vector*, $u(t) \in \mathbb{R}^m$ the *input vector* chosen by the agent, and $y(t) \in \mathbb{R}^q$ the *output vector*, i.e. the output observed and thus available to the feedback controller. Accordingly, $A_c \in \mathbb{R}^{n\times n}$ is called the *system matrix*, $B_c \in \mathbb{R}^{n\times m}$ the *input matrix*, $C \in \mathbb{R}^{q\times n}$ the *output matrix*, and $D \in \mathbb{R}^{q\times m}$ the *feedthrough matrix*.

The two equations in (2.1) play fundamentally different roles, and it is helpful to explain their distinction. Equation (2.1a) is a differential equation: it does not

compute the state, which is assumed to be known at every instant, but prescribes its *rate of change*. The pair (A_c, B_c) therefore governs how the system *behaves*, with A_c describing its autonomous dynamics and B_c describing how the input of the agent influences them. Equation (2.1b), in contrast, contains no derivative at all. It is an instantaneous readout, and the pair (C, D) therefore describes how the system is *observed* rather than how it evolves. We set $D = 0$ throughout, which is the standard assumption for systems in which the actuator has no direct instantaneous influence on the sensor[CITE: either same notes or textbook].

Given an input signal $u(\cdot)$, the trajectory of (2.1a) is obtained by integration and is given in closed form by the variation-of-constants formula

$$x(t) = e^{A_c t} x_0 + \int_0^t e^{A_c(t-s)} B_c \, u(s) \, \mathrm{d}s, \tag{2.2}$$

where $e^{A_c t}$ denotes the matrix exponential[CITE: same textbook]. This makes the interpretation of A_c precise: in the scalar case $\dot{x} = ax$ yields $x(t) = e^{at} x_0$, so a is a growth *rate* rather than a value. Equation (2.2) generalizes this, with the eigenvalues of A_c acting as growth rates along the corresponding eigendirections.

Full-state measurement. Generally, we assume that the state is fully measurable and therefore set $C = I_n$, which reduces (2.1b) to $y(t) = x(t)$. However, when we introduce disturbances through noise, this is no longer the case. More details on this are given in Section 2.3.1.2. The distinction between $u = -Kx$ and $u = -Ky$ is therefore critical, since the former admits an optimal solution while the latter might not; we return to this point in Section 2.1.4.

2.1.1.1 Discretization

Digital controllers, as in our case, do not act continuously: they sample the output at a fixed rate and hold the computed control input constant until the next sampling point. The learning algorithms studied in this thesis operate on such sampled trajectories as well, so our model has to reflect this. The standard approach is the *zero-order hold* (ZOH)[CITE: find reference]. Fixing a sampling period $h > 0$ and writing $x_k \coloneqq x(kh)$, the input is assumed to be piecewise constant,

$$u(t) = u_k \qquad \text{for } t \in [kh, \, (k+1)h), \quad k \in \mathbb{N}_0. \tag{2.3}$$

Substituting (2.3) into (2.2) over a single sampling interval and applying the change of variables $\tau = (k+1)h - s$ yields the discrete-time model

$$x_{k+1} = A_d \, x_k + B_d \, u_k, \tag{2.4}$$

with

$$A_d = e^{A_c h}, \qquad B_d = \left(\int_0^h e^{A_c \tau} \, \mathrm{d}\tau \right) B_c. \tag{2.5}$$

Two clarifications need to be made. First, (2.4) is not an approximation: it reproduces the continuous-time trajectory exactly at the sampling instants, provided the input is genuinely piecewise constant. This distinguishes ZOH from other schemes such as the forward Euler discretization $A_d \approx I + hA_c$, $B_d \approx hB_c$, which is only first-order accurate.[1] Second, the left-hand side of (2.4) is the actual next state rather than a derivative. The identity $(x_{k+1} - x_k)/h \approx \dot{x}(kh)$ recovers the continuous picture, and the identity block contained in A_d but not in A_c is exactly the state carried over from the previous step.

Unless stated otherwise, all experiments in this thesis are carried out in discrete time with sampling period $h = 0.01\,\mathrm{s}$, and we drop the subscript d in the following, writing simply A and B.

2.1.2 The Double Integrator

To make the comparisons across all chapters meaningful, all experiments are built on a single system: the *double integrator*, a frictionless point mass of unit mass moving along a line under an applied force u[CITE: find some reference]. Its dynamics are Newton's second law,

$$\ddot{p}(t) = u(t), \tag{2.6}$$

where $p(t)$ denotes the position of the mass. We simplify by using a mass of one, so that force and acceleration coincide numerically. Choosing position and velocity as the state, $x = [\,p,\ v\,]^\top \in \mathbb{R}^2$, the second-order equation (2.6) decomposes into the first-order system

$$\dot{p}(t) = v(t), \qquad \dot{v}(t) = u(t), \tag{2.7}$$

in which the first equation is the definition of velocity and the second is (2.6) itself. In the notation of (2.1) this gives

$$A_c = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \qquad B_c = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \qquad C = I_2, \qquad D = 0. \tag{2.8}$$

The entry $(A_c)_{12} = 1$ should be read as stated in Section 2.1.1: the rate of change of position is given by the current velocity.

Because $A_c^2 = 0$, the matrix exponential in (2.5) truncates after two terms, $e^{A_c h} = I + A_c h$, and the discretization is available in closed form:

$$A_d = \begin{bmatrix} 1 & h \\ 0 & 1 \end{bmatrix}, \qquad B_d = \begin{bmatrix} h^2/2 \\ h \end{bmatrix}. \tag{2.9}$$

This system is the minimal nontrivial choice for our purposes. It is second-order, so the state is not directly actuated and the controller must account for momentum; it is controllable for every $h > 0$; and it is not open-loop stable, since A_c has a double

[1] For the double integrator of Section 2.1.2 the two approaches happen to agree on A_d but *not* on B_d, where forward Euler gives $[0,\ h]^\top$ instead of the correct $[h^2/2,\ h]^\top$.

eigenvalue at the origin and A_d correspondingly a double eigenvalue at 1. The last property matters for the learning experiments: an uncontrolled or badly initialized policy diverges, so the assumption of a known stabilizing initial gain (Section 2.1.4) is a real restriction. These properties are made precise in Section 2.1.3.

2.1.3 Characteristics of a System

Now that we have defined our system, before we can optimize over it, it is important to clear up different characteristics of the same. The three we need are properties of the model and not of any particular controller. Stability is what the controller has to establish in the first place, controllability is what guarantees that it can, and observability is what allows a controller that only sees y to act as if it actually recorded x.

2.1.3.1 Stability

The first of these is stability, which guarantees that deviations in the system do not become too drastic, and thus cause some kind of failure. Mathematically, stability is a statement about a *solution* of the recursion $x_{k+1} = f(x_k, k)$, and is defined the following way [CITE: Astrom & Wittenmark, "Computer-Controlled Systems", p.77]:

Definition 2.1 (Stability)**.** Let $x^0(\cdot)$ be a solution of $x_{k+1} = f(x_k, k)$. The solution $x^0(\cdot)$ is called *stable* if for every $\varepsilon > 0$ there exists a $\delta(\varepsilon, k_0) > 0$ such that every solution $x(\cdot)$ with $\|x_{k_0} - x^0_{k_0}\| < \delta$ satisfies

$$\|x_k - x^0_k\| < \varepsilon \qquad \text{for all } k \geq k_0.$$

It is called *asymptotically stable* if it is stable and in addition $\|x_k - x^0_k\| \to 0$ as $k \to \infty$.

In the definition, ε bounds how far a perturbed trajectory is allowed to deviate, and δ says how small the initial distance must be to stay inside that bound. Asymptotic stability additionally requires the deviation to vanish with increasing time.

For linear systems this simplifies considerably. If $x(\cdot)$ and $x^0(\cdot)$ both solve $x_{k+1} = Ax_k$, then their difference solves the same recursion, so the behavior of *one* solution decides the behavior of all of them. Stability is thus a property of the system, explicitly the matrix A. This is stated as a theorem by [CITE: Astrom & Wittenmark, "Computer-Controlled Systems", p

Theorem 2.1 (Asymptotic stability of discrete-time LTI systems)**.** *The discrete-time LTI system $x_{k+1} = Ax_k$ is asymptotically stable if and only if all eigenvalues of A lie strictly inside the unit disk, i.e.*

$$\rho(A) := \max_i |\lambda_i(A)| < 1.$$

Our double integrator sits precisely on the boundary. From (2.9), A_d has the double eigenvalue $\lambda_{1,2} = 1$, so $\rho(A_d) = 1$ and Theorem 2.1 fails. It is not only

non-asymptotically stable but actually unstable. We can see that in the matrix powers

$$A_d^k = \begin{bmatrix} 1 & kh \\ 0 & 1 \end{bmatrix}, \tag{2.10}$$

which are growing without bound in k. This is also logical from a physical point of view as a frictionless mass with a nonzero initial velocity continues its trajectory forever if nothing acts on it. Every controller we construct therefore has to *create* stability rather than preserve it.

2.1.3.2 Controllability

Whether a system is controllable or not indicates if, independently of the controller, it is possible to reach any state inside its state space in a given time. A more formal definition is given in [CITE: ControlSystemsTheoryAndDesign]:

Definition 2.2 (Controllability)**.** The system is said to be *controllable* if for any initial state x_0 and any final state x_f there exist a horizon N and inputs $u_0, \dots, u_{N-1}$ such that the solution of (2.4) satisfies $x_N = x_f$. Otherwise, the system is said to be *uncontrollable*.

To check whether a given system is actually controllable, we need to compute the *controllability matrix* [CITE: ControlSystemsTheoryAndDesign]

$$\mathcal{C}(A, B) = \begin{bmatrix} B & AB & A^2B & \cdots & A^{n-1}B \end{bmatrix} \in \mathbb{R}^{n \times nm}, \tag{2.11}$$

and then verify that it has full rank n; the system is controllable if and only if it does [CITE: ControlSystemsTheoryAndDesign]. Note that (2.11) is stated for a general input matrix B; for a single input ($m = 1$) it reduces to the square matrix $\begin{bmatrix} b & Ab & \cdots & A^{n-1}b \end{bmatrix}$.

Applying this to our double integrator, we obtain

$$\mathcal{C}(A_d, B_d) = \begin{bmatrix} B_d & A_dB_d \end{bmatrix} = \begin{bmatrix} h^2/2 & 3h^2/2 \\ h & h \end{bmatrix}, \qquad \det \mathcal{C}(A_d, B_d) = -h^3, \tag{2.12}$$

which is nonzero for every sampling period $h > 0$. The system is therefore controllable at every sampling rate, as claimed in Section 2.1.2.

For Section 2.1.4 we do not need the strong Definition 2.2 but can rather suffice with the weaker notion of *stabilizability* [CITE: ControlSystemsTheoryAndDesign]:

Definition 2.3 (Stabilizability)**.** The pair (A, B) is called *stabilizable* if there exists a gain $K \in \mathbb{R}^{m \times n}$ such that $\rho(A - BK) < 1$, i.e. if the closed loop obtained from (2.4) with $u_k = -Kx_k$ can be made asymptotically stable. Equivalently, (A, B) is stabilizable if and only if every mode of A that is not controllable is already asymptotically stable.

However, to decide whether our system is stabilizable, we do not need to construct this specific K. The Popov–Belevitch–Hautus (PBH) test [CITE: TODO: Find fitting] allows us to employ it as a rank condition on the relevant eigenvalues:

Theorem 2.2 (PBH test for stabilizability)**.** *The pair (A, B) is stabilizable if and only if*

$$\operatorname{rank} \begin{bmatrix} A - \lambda I_n & B \end{bmatrix} = n \qquad \text{for all } \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1. \tag{2.13}$$

Comparing (2.13) with (2.11) shows that the latter applies the same rank condition as the former, however imposed for *every* $\lambda \in \mathbb{C}$. Thus, controllability implies stabilizability but not vice versa. The difference contains systems which have uncontrollable directions decaying on their own, and a controller can hence never influence.

As our double integrator is already controllable by (2.12), it is therefore also stabilizable.

2.1.3.3 Observability

Finally, we need to take a look at observability. Above we assumed that we can actually observe the state of the system we are modeling. However, it is much more realistic that we only collect a recording of it, for example through sensors, and need to estimate the real state of the system. The notion of observability describes whether we can determine this state from our control input u and our recorded response y alone [CITE: ControlSystemsTheoryAndDesign].

Definition 2.4 (Observability)**.** The system with state-space model (2.4) and output $y_k = Cx_k$ is said to be *observable* if the initial state x_0 can be uniquely determined from the time history of the input u_k and the output y_k over a finite interval. Otherwise, the system is said to be *unobservable*.

Dual to the controllability matrix, we can derive the *observability matrix*

$$\mathcal{O}(C, A) = \begin{bmatrix} C \\ CA \\ \vdots \\ CA^{n-1} \end{bmatrix} \in \mathbb{R}^{qn \times n}, \tag{2.14}$$

and make use of the following theorem [CITE: ControlSystemsTheoryAndDesign]:

Theorem 2.3 (Observability rank condition)**.** *The system is observable if and only if the observability matrix $\mathcal{O}(C, A)$ has full rank n.*

Using Theorem 2.3 for our double integrator is actually quite simple: we set $C = I_2$ in (2.8), so $\mathcal{O}(C, A_d)$ already contains the identity in its first block row and has full rank. The system is thus observable.

If we cannot uniquely determine x_k from our observation, we work with an estimate $\hat{x}_k$ instead. Running a copy of the model, $\hat{x}_{k+1} = A\hat{x}_k + Bu_k$, the estimation error $\tilde{x}_k \coloneqq x_k - \hat{x}_k$ follows the dynamics

$$\tilde{x}_{k+1} = A\,\tilde{x}_k, \qquad \tilde{x}_0 = x_0 - \hat{x}_0, \tag{2.15}$$

which is again just the system without control input. The decay of the error therefore depends only on the eigenvalues of A. If the system is unstable, the error grows infinitely [CITE: ControlSystemsTheoryAndDesign] and for our example it does so linearly, by (2.10). Observability fixes this by correcting the model with the measurement, $\hat{x}_{k+1} = A\hat{x}_k + Bu_k + L(y_k - C\hat{x}_k)$, which turns (2.15) into $\tilde{x}_{k+1} = (A - LC)\tilde{x}_k$. If the system is observable, we can choose an L such that the eigenvalues of $A - LC$ lie inside the unit disk, making it stable.

As with stabilizability, the weaker requirement is called *detectability*, and it is the condition we rely on in Section 2.1.4.

Definition 2.5 (Detectability)**.** The pair (A, C) is called *detectable* if there exists an observer gain $L \in \mathbb{R}^{n\times q}$ such that $\rho(A - LC) < 1$. Equivalently, (A, C) is detectable if and only if every unobservable mode of A is already asymptotically stable.

Detectability is the dual of Definition 2.3 which gives us that (A, C) is detectable if and only if $(A^\top, C^\top)$ is stabilizable [CITE: ControlSystemsTheoryAndDesign]. This follows as $\rho(A - LC) = \rho\big(A^\top - C^\top L^\top\big)$. Transposing the test for stabilizability (2.13) leaves us with the dual rank test:

Theorem 2.4 (PBH test for detectability)**.** *The pair (A, C) is detectable if and only if*

$$\operatorname{rank}\begin{bmatrix} A - \lambda I_n \\ C \end{bmatrix} = n \qquad \text{for all } \lambda \in \mathbb{C} \text{ with } |\lambda| \geq 1. \tag{2.16}$$

With $C = I_2$ the matrix in (2.16) contains an identity block and has full rank for every λ, so our system is observable and therefore detectable. The reason we introduce the weaker notion here is Theorem 2.5, whose hypothesis is the detectability of $(A, Q^{1/2})$ rather than (A, C). This shifts the focus from what the sensor sees to which directions of the state space are penalized by the cost.

The topic of observability becomes important again in Section 2.3.1.2, where we deliberately introduce noise so that $y_k \neq x_k$. This does not change our matrix $C = I_2$, so the rank condition of Theorem 2.3 still holds and the system remains observable by definition. However, y_k is no longer a deterministic function of x_k, so we can never estimate the state exactly. We revisit this and its implications there.

2.1.4 The Linear Quadratic Regulator

Now that we have defined our state-space model and its important characteristics, it is time to introduce our controller, which is based on the *linear quadratic regulator* (LQR) problem.

Here, the setting is that our system starts at an arbitrary initial state in its state space, and the goal of our controller is to return to the zero state as fast as possible. At the same time, the control input should be limited to a realistic quantity, such that it could actually be produced by an actuator. Both requirements can be combined into a single cost function, where Q describes the former, while R associates costs to control input. Over a finite horizon T, this gives the cost [CITE: OptimalAndRobustControl.pdf]

$$V_T(x_0, u) = \int_0^T \left(x(t)^\top Q\, x(t) + u(t)^\top R\, u(t)\right) \mathrm{d}t, \qquad Q = Q^\top \succeq 0, \quad R = R^\top \succ 0. \tag{2.17}$$

We notice two further restrictions here: Q is required to be positive semi-definite, meaning deviating from the origin is not reducing costs, and R must be positive definite, such that every control input results in a cost.

As the original formula relates to a continuous problem with a fixed end time T, it does not fit our goal well of evaluating over an infinite discrete horizon. The following formula offers exactly that:

$$J(x_0, u) = \sum_{k=0}^{\infty} \left(x_k^\top Q\, x_k + u_k^\top R\, u_k\right). \tag{2.18}$$

This sum does not necessarily converge. If the closed loop does not decay, the costs do not vanish and J tends to $+\infty$. Using this as our definition, J becomes well defined for every gain.

2.1.4.1 The Optimal Gain

What makes the LQR problem special is that its solution is a static linear feedback law: the optimal input at time k depends only on the current state of the model. Introducing a gain matrix $K \in \mathbb{R}^{m \times n}$, we write it as [2]

$$u_k = -K x_k. \tag{2.19}$$

Our controller is thus reduced to choosing a matrix, and the optimal controller is given by the optimal gain matrix K^*.

As we are dealing with a discrete-time system, that gain is obtained from the *discrete algebraic Riccati equation* (DARE)

$$P = A^\top P A - A^\top P B \left(R + B^\top P B\right)^{-1} B^\top P A + Q, \tag{2.20}$$

[2] Note that the minus in front of K has been chosen as the convention for this thesis. This turns our closed loop into the format $A - BK$, while some referenced literature uses $A + BK$. Formulas cited have been adapted to our convention by replacing K with $-K$.

whose stabilizing solution $P = P^\top \succeq 0$ yields

$$K^* = \left(R + B^\top P B\right)^{-1} B^\top P A, \qquad u_k^* = -K^* x_k. \tag{2.21}$$

The matrix P is additionally our *cost-to-go* matrix, meaning that we can use it to calculate the optimal cost incurred from any state x onwards with

$$J^*(x) = x^\top P x. \tag{2.22}$$

Whether (2.20) has such a solution at all is answered by the structural properties of Section 2.1.3 [CITE: OptimalAndRobustControl.pdf].

Theorem 2.5 (Existence and uniqueness)**.** *Let (A, B) be stabilizable and let $(A, Q^{1/2})$ be detectable. Then the DARE (2.20) has a unique symmetric positive semi-definite solution P, which is the only stabilizing one, and the resulting closed loop satisfies $\rho(A - BK^*) < 1$. If $(A, Q^{1/2})$ is observable, P is positive definite.*

2.1.4.2 The Cost as a Function of the Gain

Equation (2.21) solves the problem, but it solves it directly and tells us nothing about the quality of non-optimal gains. The learning algorithms of the following chapters never see the model itself; they search over gains directly, and for that we need the cost as a function of K rather than of an input sequence.

Fix a gain K and write the resulting closed loop as $x_{k+1} = A_K x_k$ with $A_K := A - BK$. Substituting (2.19) into (2.18) shows that the cost is again a quadratic form, $J(x_0, K) = x_0^\top P_K x_0$, where P_K now solves the far simpler discrete *Lyapunov equation*

$$P_K = A_K^\top P_K A_K + Q + K^\top R K, \tag{2.23}$$

which is linear in the unknown P_K and has a unique solution whenever $\rho(A_K) < 1$. This restricts us to the set of stabilizing gains

$$\mathcal{K} := \left\{K \in \mathbb{R}^{m \times n} : \rho(A - BK) < 1\right\}, \tag{2.24}$$

outside of which we set the cost to $+\infty$, consistent with the remark after (2.18).

One dependency remains, namely on the initial state. Evaluating the gain based on a single x_0 would reward gains that happen to fit that one point, so we instead draw the initial state from a distribution, $x_0 \sim \mathcal{D}$, and average. Writing $\Sigma_0 := \mathbb{E}_{x_0 \sim \mathcal{D}}[x_0 x_0^\top]$ for its second moment, the objective becomes

$$f(K) := \mathbb{E}_{x_0 \sim \mathcal{D}}\left[x_0^\top P_K x_0\right] = \operatorname{tr}\left(P_K \Sigma_0\right), \tag{2.25}$$

using the cyclic property of the trace. Equation (2.25) is the object that every remaining chapter of this thesis minimizes, and the algorithms differ only in how much of the model is known to them.

2.2 Motivation

An agent acting in its environment receives observational input and responds to it with a commanded control input. In other words, the agent is a function in relation to its environment, and to find an optimal behavior, we need to approximate this optimal function as closely as possible. While neural networks are known to be universal approximators [4, 13], and hence able to approximate any function, in the case of an embodied agent, different constraints need to be taken into account. Ay [2] names additionally to external embodiment constraints possible intrinsic constraints, and argues that such can even be "beneficial" for the control structure. Additionally, fewer parameters for deployment are helpful due to energy usage and storage capacity, the core of the notion of cheap design[CITE: TODO]. So, for such embodied agents, the goal is to find an "embodied universal approximator" [27, p. 3], which restricts the neural network architecture to an intrinsic, unknown dimension, encoding physical properties belonging to the agent and the environment, while disallowing impossible movements [27, Sec. 3].

Biology motivates this approach even further: here it is clear to see that by commanding a body, some features of the control architecture can be simplified and delegated to the embodiment[CITE: TODO]. For example, McGeer's passive walker [25] clearly shows that some movements can be produced by the body without any control input, just by choosing the right embodiment. While obviously here being simplified to not include any control at all, it can quite simply be extended with such[CITE: #9, McGeer reference to his previous paper]. Müller and Hoffmann [28, p. 4] mention such a combination, a robot climbing up walls, whose design was inspired by geckos, indeed using energy for control, however very little compared to an approach not exploiting this specific embodiment design. Further, it can be seen that the complex behaviors are not only less energy-intensive to produce, but also utilize muscle synergies [6] and thus fewer control parameters. And while it is difficult to clearly attribute the effect of the embodiment [49], for us, this gives reason to believe that the simplicity, and thus efficiency, comes from a low-dimensionality of the controller [27].

That this is indeed true for certain examples has been shown both in theory and practice before [5, 26, 23]. Li et al. [23] call this lower dimension "intrinsic dimension", which is the dimensionality required for the model to be able to find policies. They show, for example, that for achieving 90 % of the performance of a fully connected network on MNIST, the dimensionality can be reduced by 99.6 %. For a convolutional neural network, it is still a 99.3 % reduction. Others show that the "least complex model which is able to learn an optimal policy does learn faster" [26]. For a neural network, reducing parameters can be done in different ways[CITE: TODO], some of which we look at in Chapter 4. In comparison, for LQR, where the controller is a gain matrix, the corresponding approach is reducing its rank [34]. However, while these approaches are more parameter-efficient while achieving similar or slightly worse performance, it was also observed that they tend to perform even worse when encountering noise [5].

Robustness to noise, however, is one of the important characteristics of controllers acting under real-world conditions [32, 35]. One approach of achieving this in nature is the principle of "redundancy and diversity" [19], which stands in direct contrast to our goal of reducing redundant parameterizations. Another reason for producing robust controllers is the difficult transfer from digital training environment to real-world embodiments, commonly referred to as the "reality gap" [44, 30]. Approaches of training robust controllers typically introduce noise already during training, be it either by using a linear quadratic Gaussian (LQG) controller or by including variations to the dynamics [32, 30]. These approaches, however, do not look at low-dimensionality, which makes sense as Kitano [19, p. 833] poses that robustness cannot be increased without sacrificing performance or accepting "increased resource demands".

Before taking a look at the more complex case of RL, we start simpler by using the simpler and better-understood LQR. This approach has been taken before [33] and does make sense given the guarantees regarding convergence and effectiveness of policy optimization for it [12, 42], as well as the understanding of its geometry [43]. It also offers methods which produce certain structures in its controller, like sparsity or our desired low-dimensionality, as given by structured policy iteration (S-PI) [29]. Approaches can generally be categorized into two groups if the underlying dynamics are unknown and only data is available: estimating the model and then applying a model-based solver (indirect), or using the data to estimate the gain matrix (direct) [51]. Both the indirect, as in system identification with uncertainty bounds [7], and the direct setting, with an $\mathcal{H}_\infty$ constraint that bounds the closed-loop gain from disturbance to output [50], have been studied with regard to robustness, however, to our knowledge, not in combination with low-rank policies.

Research evaluating RL on control tasks has also shown that there is room for simplification, for example using linear policies instead of complex, fully connected neural networks [24, 31]. This, together with the fact that parameters in neural networks are highly dependent, and thus redundant [9], allows us to assume that low-dimensionality can improve efficiency here. Examples of this exist, for example by factorizing a discretized low-rank policy into a thin and wide matrix [36] or by introducing an orthogonal projection inside the layer structure [45]. While they produce lower dimensionality, a different approach uses "information bottlenecks", which exhibit better generalization behavior in unseen environments [16, Sec. 4.2], which again could help us with robustness.

A clear case where low-dimensional dynamics underlie a high-dimensional observational space is in perception-based tasks. Agents observing only images instead of the states of simple control tasks are posed with the challenge of extracting these states from a high-dimensional, often nonlinear encoding [47, Sec. 1]. Many approaches, based on state representation learning (SRL), exist, which use different variants to create a latent embedding of the image, used as a state [21]. The most interesting approach, especially for us looking at embodied agents, is the usage of

so-called "priors", which restrict the latent space by using knowledge about the environment [21]. This brings us back to Ay's argument that the embodiment actively defines the action space [2]. Robotic priors were introduced by Jonschkowski and Brock [17] in 2015, and then further refined by Botteghi et al. [3] for continuous action spaces. A more specific version restricted to position and velocity also exists here, training a model to encode an image into a low-dimensional position state, from which the velocity is calculated by finite differences [18]. While all that sounds promising, in contrast stand findings hinting that the problems of robust control also apply to perception-based control, as noise induced through perception directly affects the performance [48] and thus needs approaches for improving it [8].

Our goal for this thesis is therefore to look a bit deeper at the combination of low-dimensionality and robustness in an empirical way, motivated by the findings of Datar and Ay [5]. LQR serves as the perfect, well-investigated basis [33], offering low-rank approaches that have been proposed and analyzed theoretically, but have not yet been applied in practice and under noise [29]. Previously investigated policies for LQR under noise in comparison did not look at low-dimensionality specifically [50]. For RL, low-dimensionality has been explored previously [1, 41], and also in comparison with robustness [22], but does not take into account the embodiment we are interested in. Closing this gap, our main interest is to compare robustness properties for low-dimensional policies for embodied agents under increasingly more complex and noisier environments.

2.3 Experimental Setup

2.3.1 The Modified Loop

2.3.1.1 An Inflated Model

To allow us to investigate a controller that has many redundant inputs and outputs, acting on a system with low-dimensional dynamics, we first need to create such a setup for our previously introduced double integrator. We do this by keeping the dynamics, so the state progression, like the loop above, but projecting the observation into a higher dimension, as well as projecting actuation down from such a higher dimension.

More specifically, we show the modification to our loop with two random orthonormal matrices U_1 and U_2. For an inflated input dimension d_{in} and output dimension d_{out}, the two lifts are constructed the following way:

$$U_1 \in \mathbb{R}^{d_{\text{in}} \times n}, \quad U_1^\top U_1 = I_n, \qquad U_2 \in \mathbb{R}^{m \times d_{\text{out}}}, \quad U_2 U_2^\top = I_m, \tag{2.26}$$

where U_1 is the orthonormal factor of the QR decomposition of a random matrix, whose entries are sampled from the standard normal distribution. U_2 is the transpose of such a factor sampled from a separately drawn matrix. A single step of

the loop then becomes

$$
\begin{aligned}
y_k &= U_1 x_k, && \text{(2.27a)}\\
v_k &= -K y_k, \qquad K \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}, && \text{(2.27b)}\\
u_k &= U_2 v_k, && \text{(2.27c)}\\
x_{k+1} &= A x_k + B u_k = A x_k + B U_2 \, v_k. && \text{(2.27d)}
\end{aligned}
$$

Compared with (2.1), the output matrix thus becomes $C = U_1$, so that $q = d_{\text{in}}$, and the controller acts through the input matrix $BU_2 \in \mathbb{R}^{n \times d_{\text{out}}}$.

This way, the controller only sees the inflated observation and acts in a higher dimension. As any actuation effort is penalized, R acts on the inflated actuation and therefore has the dimension $d_{\text{out}} \times d_{\text{out}}$. To simplify things, we also have Q in this lifted dimension, acting on the observation. However, as U_1 has orthonormal columns, and we use $Q = I_{d_{\text{in}}}$ throughout, this simplifies to

$$
\begin{aligned}
(U_1 x_k)^\top Q \, (U_1 x_k) &= x_k^\top U_1^\top U_1 \, x_k \\
&= x_k^\top I_n \, x_k \\
&= x_k^\top Q_n \, x_k, \qquad Q_n \coloneqq I_n \in \mathbb{R}^{n \times n},
\end{aligned}
\tag{2.28}
$$

so it indeed still penalizes how far our mass is from zero. The cost in the modified loop then becomes

$$
\begin{aligned}
J(x_0, K) &= \sum_{k=0}^{\infty} \left(y_k^\top Q \, y_k + v_k^\top R \, v_k \right) \\
&= \sum_{k=0}^{\infty} \left(x_k^\top U_1^\top Q \, U_1 x_k + x_k^\top U_1^\top K^\top R K U_1 x_k \right),
\end{aligned}
\tag{2.29}
$$

with the closed loop $x_{k+1} = (A - BU_2 K U_1) \, x_k$.

Throughout the thesis, we will use the following synonyms for different d_{in} and d_{out} dimensionalities, applied to the loop, which can be found in Table 2.1.

Table 2.1: Preset inflation sizes.

Name	d_{in}	d_{out}
base	2	1
small	10	5
medium	50	25
large	200	100

2.3.1.2 Noise Injection

Now, the second thing we want to take a look at is robustness to small changes or variations in the environment. We thus introduce four different types of disturbances to be included in our model, commonly known as noise. They are distinguished by the point at which they affect the loop: what the controller *observes* (measurement noise), what its output actually *does* (control noise), what the plant *is* (parametric uncertainty), and what else *acts* on it (external disturbance). Each of them can be activated separately, so that we can clearly isolate the cause of an effect. Throughout this section k indexes the sampling instants, and σ denotes the standard deviation of the respective noise.

2.3.1.3 Measurement Noise

Here we influence the sensors used to measure the model state, such that capturing the same state twice does not lead to the same observation y_k.

Let n_k^{m} denote the measurement noise at step k. Then

$$n_k^{\mathrm{m}} \sim \mathcal{N}\big(0, \sigma_{\mathrm{m}}^2 I_{d_{\mathrm{in}}}\big)\,, \qquad y_k = U_1 x_k + n_k^{\mathrm{m}}, \qquad \sigma_{\mathrm{m}} \in \mathbb{R}_+, \tag{2.30}$$

drawn independently at every step.

2.3.1.4 Control Noise

Control noise, as the name says, acts on the applied control input. With it, we try to model the difference between the intended movement of an actuator and its actual action due to different factors. Here we deal with *multiplicative* noise, which scales with the input. In detail this means that the applied controller output $\tilde{v}_k$ is obtained as

$$n_k^{\mathrm{c}} \sim \mathcal{N}\big(0, \sigma_{\mathrm{c}}^2 I_{d_{\mathrm{out}}}\big)\,, \qquad \tilde{v}_k = (\mathbf{1}_{d_{\mathrm{out}}} + n_k^{\mathrm{c}}) \odot v_k, \tag{2.31}$$

where $\odot$ denotes the elementwise product, so that σ_{c} is the *relative* error. Applying this to the input, the next step becomes

$$x_{k+1} = A\,x_k + BU_2\,\tilde{v}_k. \tag{2.32}$$

2.3.1.5 Parametric Uncertainty

This noise acts directly on the matrices of the model. Here, we either decide to keep it time invariant, meaning $\Delta_k^A = \Delta_0^A$ and $\Delta_k^B = \Delta_0^B$ for all k, or to resample it at every step. The former models a wrongly identified system, while the latter simulates one that continuously changes. As with measurement noise, it is additive. Applied, it acts on the system the following way:

$$\big(\Delta_k^A\big)_{ij}\,,\ \big(\Delta_k^B\big)_{ij} \overset{\text{i.i.d.}}{\sim} \mathcal{N}\big(0, \sigma_{\mathrm{p}}^2\big)\,, \qquad x_{k+1} = \big(A + \Delta_k^A\big)\,x_k + \big(B + \Delta_k^B\big)\,U_2\,\tilde{v}_k. \tag{2.33}$$

Worth noting is that resampling every step leaves the model correct *on average*, since $\mathbb{E}[\Delta_k^A] = \mathbb{E}[\Delta_k^B] = 0$, and mainly adds variance to the trajectory. This stands in contrast to the time-invariant version, where we have a discrepancy between what we compute actions on, and what actually happens.

2.3.1.6 External Disturbance

This noise models external influences, like a wind pushing on our agent, or rain hitting it with some frequency. It only affects the position, so with the first unit vector $e_1 \in \mathbb{R}^n$ the noise is drawn as

$$n_k^{\mathrm{d}} = \sigma_{\mathrm{d}} \sin(0.01\, k)\, e_1. \tag{2.34}$$

As the disturbance acts on the true state x_k, it is applied before any lifting of the observation (2.27a). The loop can thus be described as

$$x_{k+1} = A\, x_k + B U_2\, \tilde{v}_k + n_k^{\mathrm{d}}. \tag{2.35}$$

2.3.1.7 The Full Loop

While we, until now, have only regarded each noise type on its own, there is also the possibility to include them all at once. Combining all the channels with the inflated loop, a single step becomes

$$\begin{aligned} y_k &= U_1 x_k + n_k^{\mathrm{m}}, && \text{(2.36a)} \\ v_k &= -K y_k, && \text{(2.36b)} \\ \tilde{v}_k &= (\mathbf{1}_{d_{\mathrm{out}}} + n_k^{\mathrm{c}}) \odot v_k, && \text{(2.36c)} \\ x_{k+1} &= \left(A + \Delta_k^A\right) x_k + \left(B + \Delta_k^B\right) U_2\, \tilde{v}_k + n_k^{\mathrm{d}}. && \text{(2.36d)} \end{aligned}$$

2.3.2 Metrics and Robustness

In this thesis we try to answer two main questions. First, how well a trained low-dimensional controller performs generally, as well as under influences unknown to it during training. Thus we aim to define here the metrics to quantify our results.

2.3.2.1 Cost and Suboptimality

By Theorem 2.5 no gain achieves a lower cost (2.25) than K^*, and every controller we train is therefore in the best case only as good as it. This serves as our baseline, and we evaluate the cost produced by our controllers in relation to it.

While we can easily calculate the infinite-horizon cost of a gain via the Lyapunov equation (2.23), we cannot effectively use it. The reason for that is that (2.23) can only be applied to the closed loop, however when adding noise to our evaluation, this is not the case anymore. Additionally, the neural networks we train in later chapters do not admit a gain-matrix-based policy P_K, which would need a different approach anyway.

So, we instead evaluate them over a fixed horizon via rollout. Hence for a given controller, a noise condition c, which is a subset of channels and an intensity

(Section 2.3.2.3), and a horizon T, we simulate starting from a randomly drawn $x_0 \sim \mathcal{D}$. The resulting loop (2.36) under c thus accumulates

$$J_T^{(i,s)}(K;c) = \sum_{k=0}^{T-1} \left((U_1 x_k)^\top Q\,(U_1 x_k) + \tilde{v}_k^\top R\,\tilde{v}_k \right), \tag{2.37}$$

where s indexes the seed the run was trained under and i the episode within it.

It is important to note where Q and R act. As Q penalizes the difference from the origin, we apply it on the inflated state *before* any noise is applied. This way we can evaluate the actual position of the mass, as shown in (2.28). For R, we handle it differently. Here, every actuation energy adds to the cost, irrespective of its origin (so whether the controller commanded it or it is noise).

Writing $\mathcal{F}$ for the pairs (s,i) whose episode stayed finite, the cost estimate becomes

$$\hat{J}_T(K;c) = \frac{1}{|\mathcal{F}|} \sum_{(s,i)\in\mathcal{F}} J_T^{(i,s)}(K;c), \qquad \mathcal{F} = \left\{ (s,i) \;:\; J_T^{(i,s)}(K;c) < \infty \right\}. \tag{2.38}$$

Given this information, we can state our main metric, which is the relative excess of the cost produced by a trained controller over the cost of the optimal controller K^* under the same condition. It is taken *per episode*, meaning an episode (s,i) of a trained controller and episode (s,i) of the baseline are compared here. Both start from the same x_0 and are affected by the same noise. Overall this yields

$$S^{(i,s)}(K;c) = \frac{J_T^{(i,s)}(K;c) - J_T^{(i,s)}(K^*;c)}{J_T^{(i,s)}(K^*;c)}, \qquad \hat{S}(K;c) = \frac{1}{|\mathcal{F}|} \sum_{(s,i)\in\mathcal{F}} S^{(i,s)}(K;c), \tag{2.39}$$

where $S = 0$ means that the trained controller performed as well as our baseline and $S = 0.1$ that it did 10 % worse.

Note that this metric can indeed be negative. K^* is only optimal for the nominal problem (Section 2.1.4), which has exact state feedback on known matrices and no external disturbance. These conditions do not hold, unless no noise is applied, which is only true if $c = \varnothing$.

2.3.2.2 The Size of a Policy

We have often mentioned "low-dimensional" controllers until now; however, we want to make the number of parameters comparable, not only between different LQR controllers, but also across LQR and RL. So it is indeed a question of size. For a neural network the known way is simply to count the number of weights and biases of the actor [CITE: TODO]. For a gain matrix however, we need to find a definition that is comparable to it.

The simplest approach for a $K \in \mathbb{R}^{m\times n}$ is the number of its entries, mn. In Chapter 3, however, we reduce the rank of K. Optimizing the storage size for K

of rank r, the singular value decomposition [CITE: SVD_Notes.pdf] separates it into

$$K = U\Sigma V^\top, \qquad U \in \mathbb{R}^{m\times r}, \quad \Sigma = \operatorname{diag}(\sigma_1, \dots, \sigma_r), \quad V \in \mathbb{R}^{n\times r}, \tag{2.40}$$

with U and V having orthonormal columns and $\sigma_1 \geq \cdots \geq \sigma_r > 0$ being the non-zero singular values. Hence, we only need to store mr entries for U, r for the diagonal of Σ and rn for V, which in total yields $r(m+n+1)$. For a small enough r this is less, so we take the minimum of both:

Definition 2.6 (Policy size)**.** The size of a linear policy $K \in \mathbb{R}^{m\times n}$ of rank $r = \operatorname{rank}(K)$ is

$$|K| \coloneqq \min\{mn,\ r\,(m+n+1)\},$$

the number of real numbers required to store it. The size of a neural policy is its total number of trainable parameters in the actor.

2.3.2.3 Noise Intensity

Section 2.3.1.2 introduced four channels which change the system compared to the base model. Each noise channel is separated further into five intensities,

$$\textit{none} \prec \textit{weak} \prec \textit{standard} \prec \textit{strong} \prec \textit{divergent},$$

where *none* leaves the system as it is and thus serves as our control condition. All the other intensities are calibrated by how much they increase the cost of the optimal gain K^* on the uninflated system. For each channel c and intensity ℓ, we choose the standard deviation in a way such that with the test protocol of Section 2.3.3

$$\hat{J}_T(K^*; c_\ell) = \kappa_\ell\, \hat{J}_T(K^*; \varnothing), \qquad \kappa_\ell \in \{1.2,\ 1.5,\ 3,\ 10\} \tag{2.41}$$

applies for *weak*, *standard*, *strong* and *divergent* respectively. For K^*, the increase stays the same under inflation for all channels except control noise. As it acts elementwise on the inflated output, it is averaged out over the number of actuators: *strong* control noise raises the cost of K^* by a factor of 3.0 without inflation, but only by 1.22 and 1.05 at the sizes $d_{\text{in}} \times d_{\text{out}} = 50\times 25$ and 200×100. For simplicity, however, we do not vary the levels by size, but instead accept this as a side effect. The concrete standard deviations per channel are listed in Table A.2.

2.3.3 Test Protocol

Having introduced our metrics, we now define a protocol for how we train and compare the different solver approaches with these metrics.

First, we note that we will always use $Q = I_{d_{\text{in}}}$ and $R = I_{d_{\text{out}}}$, which weighs position, velocity and control effort equally. Initial states are drawn uniformly from the box $[-10, 10] \times [-1, 1]$. Some solvers require an initial stable gain, for which we will use $K_0 = [\,0.1,\ 0.5\,]$ throughout, whose closed loop has $\rho(A - BK_0) \approx 0.9975$.

We train for each algorithm ten different solvers, starting from ten different fixed seeds, until convergence. We define convergence for LQR solvers as follows:

Definition 2.7 (Convergence)**.** Let g_j be the signal that a solver observes at iteration j, and let $\bar{g}_j$ be its mean over the last N_w iterations. The solver has *converged* at the first iteration j for which

$$\bar{g}_j \geq (1 - \varepsilon_\mathrm{rel})\, \bar{g}_{j-N_\mathrm{p}},$$

i.e. once the average has not decreased by the relative tolerance ε_rel within the last N_p iterations. After an iteration with an infinite signal, the mean calculation restarts from afterwards.

The signal is something the solver itself observes and optimizes for. For Model-based PG and N4SID + Model-based PG it is the cost (2.25) on the true or identified model, for S-PI and N4SID + S-PI the regularized objective of S-PI (Section 3.1), and for Model-free NPG and S-PI (model-free) the rollout estimate of the cost. All variants of S-PI additionally stop once $\|K_{j+1} - K_j\|_F < 10^{-8}$. For all of them, we use the best iterate by $\bar{g}_j$ to identify the policy we want to use.

The parameters of Definition 2.7 for the model-based and the model-free solvers are listed in Table 2.2.

Table 2.2: Parameters of the convergence rule (Definition 2.7) for the LQR solvers.

Parameter	Model-based solvers	Model-free solvers
Window N_w	1	50
Patience N_p	50	100
Relative tolerance ε_rel	10^{-4}	10^{-3}

The number of steps the plant progressed and solver iterations until convergence are also counted and reported. An upper cap of 20 000 iterations is set, to prevent looping.

For RL-based algorithms, the convergence criteria have been calibrated based on experiments which can be found in the appendix [CITE: cref, TODO]. We apply Definition 2.7 with one iteration corresponding to 25 000 environment steps. The signal g_j is the mean (scaled) cost of the training episodes completed since the previous check. As a safety limit, we use the environment step caps in Table 2.3.

Table 2.3: Safety cap on the environment steps of the RL solvers per inflation size.

	small	*medium*	*large*
Safety limit (environment steps)	10^7	$1.5 \cdot 10^7$	$3 \cdot 10^7$

After our controllers have been trained, we evaluate them per seed for *100 episodes* each, running for *2000 steps* per episode at a sampling rate of $h = 0.01\,\mathrm{s}$. Each episode calculates its empirical cost (2.37), and thus its suboptimality (2.39) in comparison to the optimal gain on the same exact trajectory, with the same sampled $x_0 \sim \mathcal{D}$. In the end, for each solver architecture, we get a distribution of suboptimalities over a total of 1000 trajectories.

The inflated size will always be the same between training and evaluation, however, for noise experiments, the controller is trained on a clean system, and then exposed to one with noise activated during testing.

Under the stronger conditions individual rollouts might diverge. To keep our mean finite, we remove them from the mean calculation, and report them separately as the divergence rate

$$D(K;c) \coloneqq \frac{1}{MN}\left|\left\{(s,i) \;:\; J_T^{(i,s)}(K;c) = \infty\right\}\right|, \tag{2.42}$$

where $M = 10$ is the number of seeds and $N = 100$ the number of episodes per seed. An individual episode can be missing from the result due to another reason. Where the *baseline* evaluation of episode (s,i) diverged, $S^{(i,s)}$ has no value at all, since (2.39) would divide by an infinity. Those episodes are excluded from $\hat{S}$ and noted separately.

2.4 Approach

As mentioned before, the LQR serves as the perfect starting example, which is exactly why we start our investigation here. For our investigations, we will use the double integrator, as mentioned in Section 2.1.2.

First, as a basis, we will start by comparing the standard policy gradient (PG) approach [12] with S-PI [29] on the true model. Their relative performance will be evaluated

Afterwards, we will take a look at the cases where we do not have access to the real model. We compare the relative performance of both direct and indirect approaches to our baseline, which is the optimal controller calculated via the DARE (2.20) on the true model. For the indirect approaches, we will use numerical algorithms for subspace state space system identification (N4SID) [46] to estimate the underlying model from one trajectory. A more detailed description, as well as the exact parameters selected, will be provided in the sections discussing the results. With this estimated model as the basis, we employ three different approaches on how to learn our policy, called N4SID + DARE, N4SID + Model-based PG and N4SID + S-PI. All these will also be discussed in Chapter 3.

Our first approach for Chapter 3 is to compare our selected methods across the inflation sizes, and see how increasing the dimension affects the performance of each. As the next step, we will introduce the different noise channels named in

Section 2.3.1.2 first to the base double integrator, where no overparameterization exists. The results we observe here will then be compared with evaluations of the controllers for a system with *large* inflation under different noise channels and strengths.

In Chapter 4, we will then first introduce RL-based approaches. With proximal policy optimization (PPO)[CITE: TODO] being one of the most used variants in robotics[CITE: TODO], we choose this as our main architecture, and apply different variations of it. First, by doing xyz[CITE: TODO, missing text] we force the trained controller to be linear, which allows us to extract it as a gain matrix that we can closely compare with our LQR approaches (Linear PPO). The fully connected, standard version is named Fully-connected PPO, and will also be evaluated. Additionally, we will introduce structures that force the observations through a low-dimensional latent space. Here, we will reuse the N4SID system identifier from before, by creating an encoding matrix from the singular value elimination, that projects the high-dimensional observation first into the low-dimensionality, and then trains PPO with this as the input. Therefore, the name for it is State encoder + PPO. A similar approach is used for Scalable encoder + PPO, which is based on Jonschkowski and Brock [17] and Botteghi et al. [3], and uses robotic priors to derive a low-dimensional latent state via a trainable encoder. Again, the construction following this encoder is PPO. Lastly, we implement the orthonormal bottlenecks introduced by Todorov et al. [45] (Bottleneck PPO). It consists of sampling a fixed matrix U_b with d_b orthonormal columns, which is added at the end of a trainable feature extractor, and used to project the output into a d_b-dimensional space. Like with the other encoders, we follow it with our PPO architecture.

All these low-dimensional approaches, if visualized, exhibit a "bow-tie" shape, which in nature can actually improve robustness [19]. So this is indeed interesting to investigate. However, first, like in Chapter 3, we start by comparing these different approaches across the different inflation sizes, without any noise applied, to see how they generally behave. Then, we introduce noise, and compare our approaches under *large* inflation dimensions.

Taking our results from the experiments on the artificially inflated dimensions, we create a new setup, still using the double integrator as our underlying structure. Here, we will observe the mass via a simulated camera, and place a number of actuators evenly distributed among angles on a half-circle. This will give us an actual high-dimensional input, as well as a high-dimensional output, that is in reality governed by low-dimensional dynamics.

By inputting images into our existing solvers, the dimensionality increases drastically. To be able to cleanly separate effects caused by the nonlinearity of the image construction from architecture-related effects, we will first artificially inflate our inputs to exactly the size that the images will be. Also, we set the output dimension to xyz[CITE: TODO], mimicking the setting we will use for the pixelated environment. This evaluation will only be done on Scalable encoder + PPO, Fully-connected PPO and Bottleneck PPO, as the other approaches either do not scale

well or are not relevant to the comparison.

Then, using the same architectural constructions for these solvers, but applied to our new environment, we compare them with the additionally introduced position-velocity encoder [18] (Position-Velocity Encoder + PPO), which uses similar robotic priors to Scalable encoder + PPO, but focuses more on position, velocity and acceleration. It learns positional encodings from the observations, and then calculates the velocity state based on finite differences. Here, it is quite clear that we use knowledge of the limitations of the embodiment to construct more optimal controllers, again in a bow-tie shape.

Chapter 3

Robustness of LQR Solvers in Different Dimensions

3.1 A Known Model

We start by comparing two approaches that have knowledge of the underlying model, and thus do not need to estimate gradients but can calculate them exactly. This allows us to clearly separate the benefits of a low-dimensional controller from estimation errors.

3.1.1 Setting

Fazel et al. [12] introduced global convergence guarantees for policy gradient methods based on the true model in 2018. The paper describes a method for calculating the gradient of a policy from the system and cost matrices, and updating the policy accordingly. As the paper uses the same $u_k = -Kx_k$ convention as we do, we can implement the loop directly in the following way.

Starting from an initial, stabilizing gain $K_0 \in \mathcal{K}$ and the second moment Σ_0 of the initial state distribution, every iteration j first solves the discrete Lyapunov equation (2.23) for P_{K_j} using the implementation from SciPy [CITE: scipy]. The state correlation matrix Σ_{K_j} then follows from solving a second Lyapunov equation,

$$\Sigma_{K_j} = (A - BK_j)\Sigma_{K_j}(A - BK_j)^\top + \Sigma_0, \tag{3.1}$$

where, for initial states drawn uniformly from $[-10, 10] \times [-1, 1]$ as described in Section 2.1.4.2, $\Sigma_0 = \operatorname{diag}(10^2/3,\ 1^2/3)$. The policy gradient

$$\nabla f(K_j) = 2\left(\left(R + B^\top P_{K_j} B\right) K_j - B^\top P_{K_j} A\right) \Sigma_{K_j} \tag{3.2}$$

is then used for a gradient step on the gain matrix. Algorithm 1 summarizes the loop.

Algorithm 1 Model-based PG

Require: stabilizing gain $K_0 \in \mathcal{K}$, second moment Σ_0, step size η

1: **for** $j = 0, 1, 2, \dots$ **do**
2: solve (2.23) for P_{K_j}
3: solve (3.1) for Σ_{K_j}
4: compute the policy gradient $\nabla f(K_j)$ by (3.2)
5: update the gain by $K_{j+1} = K_j - \eta \nabla f(K_j)$
6: **end for**

The fixed learning rate η is a hyperparameter, which we set to $\eta = 10^{-5}$ for all inflation sizes, based on experiments whose results can be found in Section C.1. This policy gradient approach serves us as the basic, fully parameterized implementation for this experiment.

It is important to note that the above implementation is described for the uninflated loop. However, this can be easily applied to our inflated version in which $\tilde{A} = U_1 A U_1^\top$, $\tilde{B} = U_1 B U_2$, as well as our inflated weights Q, R in (2.29) and $\tilde{\Sigma}_0 = U_1 \Sigma_0 U_1^\top$, such that the resulting gain is $K \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$.

Now we also want to implement a solver that produces a low-dimensional controller. As in our case the controller is characterized by the gain matrix, and for matrices dimensionality is "often [...] expressed by means of the rank" [34], we aim at learning a low-rank gain matrix. However, to make this comparable later to RL approaches that rely on the number of parameters, we measure the complexity of a gain by its policy size $|K|$ from Definition 2.6, which uses the more efficient version between the full and the factored parameterization.

Recht et al. [34] have already shown that nuclear norm minimization introduces low-rank solutions for convex matrix problems. However, as we are dealing with a non-convex optimization problem [12], we cannot transfer the results over directly. Here, Park et al. [29] offer us an approach to impose different structures on our policy during learning via regularizers based on norms, one of them being the nuclear norm. We implement the model-based approach introduced there with small adaptations in the following way.

We reuse the computation of the gradient $\nabla f(K_j)$ from Algorithm 1. For inducing the structure, we optimize the regularized cost $f(K) + \lambda \|K\|_*$, where the nuclear norm $\|\cdot\|_*$ can be reduced by eliminating singular values of K, and λ is its weight in the whole cost. It further uses an adaptive learning rate, combined with a line search criterion, consisting of a stability and a sufficient decrease condition. Starting from the initial step size $\eta = 1/(2\lambda)$, following the paper's recommendation of $\mathcal{O}(1/\lambda)$, each iteration takes a gradient step and applies the proximal operator of the nuclear norm, which soft-thresholds the singular values,

$$K^+ = U \operatorname{diag}\left(\max\{\sigma_1 - \eta\lambda, 0\}, \dots, \max\{\sigma_{\bar{r}} - \eta\lambda, 0\}\right) V^\top, \qquad \bar{r} \coloneqq \min\{d_{\text{out}}, d_{\text{in}}\}, \tag{3.3}$$

where $U \operatorname{diag}(\sigma_1, \ldots, \sigma_{\bar{r}}) V^\top$ is the singular value decomposition of the gradient step $K_j - \eta \nabla f(K_j)$. The candidate K^+ is accepted if it is stabilizing, $\rho(A - BK^+) < 1$, and satisfies the sufficient decrease condition

$$f(K^+) \leq f(K_j) - \eta \operatorname{tr}\left(\nabla f(K_j)^\top G\right) + \frac{\eta}{2}\|G\|_F^2, \qquad G \coloneqq \frac{1}{\eta}\left(K_j - K^+\right). \tag{3.4}$$

Otherwise, the step size is reduced to $\beta\eta$ and the proximal step is repeated. The learning loop further can be stopped earlier if the criterion $\|K_{j+1} - K_j\|_F \leq \varepsilon$ is satisfied. Algorithm 2 summarizes the loop.

Algorithm 2 S-PI

Require: stabilizing gain $K_0 \in \mathcal{K}$, second moment Σ_0, weight λ, factor β, tolerance ε

1: **for** $j = 0, 1, 2, \ldots$ **do**
2: compute $\nabla f(K_j)$ as in Algorithm 1
3: set the initial step size $\eta = 1/(2\lambda)$
4: **for** at most 20 line search steps **do**
5: compute the candidate K^+ by (3.3)
6: **if** $\rho(A - BK^+) < 1$ and (3.4) holds **then**
7: **break**
8: **end if**
9: reduce the step size, $\eta \leftarrow \beta\eta$
10: **end for**
11: update the gain, $K_{j+1} = K^+$
12: **if** $\|K_{j+1} - K_j\|_F \leq \varepsilon$ **then**
13: **stop**
14: **end if**
15: **end for**

As one can see, there are a lot of parameters that can be modified here. We set $\varepsilon = 10^{-8}$, and use the same $\beta = 0.5$ as in the experiments section of Park et al. [29]. For λ, as it defines the learning rate as well, we have run an experiment with various values, which can be found in Section C.2. A value of $\lambda = 5$ has shown the most promising results across the sizes.

One thing that we implemented differently than the paper is the cap on the number of line search iterations in line four, which we set to 20. Some of our previous experiments had shown that if we leave it uncapped, for unstable updates which are impossible to stabilize, the loop runs until the update produces no effect at all.

Now that we have defined our controllers and how they learn, we test them in comparison to the DARE-calculated solution (2.20) from Section 2.1.4.1.

3.1.2 Results

We apply our solvers to the *base* and inflated versions of the true model, for now noise-free, as described in Section 2.3.1.1. Specifically, as in Section 2.3.3, we train ten controllers per solver, before evaluating them on the test episodes. The results are then pooled.

The most interesting finding is how much faster S-PI converges compared to Model-based PG. While it achieves its best training suboptimality within around 50 iterations, Model-based PG still falls drastically short of it after 450 iterations. We can also see in Table 3.1 that the reduction in complexity for all inflation sizes has worked well with S-PI. For our test suboptimality, S-PI also outperforms Model-based PG here by at least one order of magnitude. Figure 3.1 visualizes this.

Table 3.1: Size of the policy architectures on the true model, across every inflation size.

	Policy parameters				Gain rank			
Solver	base	small	medium	large	base	small	medium	large
Baseline	2	16	76	301	1.0	1.0	1.0	1.0
Model-based PG	2	50	1250	20000	1.0	5.0	25.0	100.0
S-PI	2	16	76	301	1.0	1.0	1.0	1.0

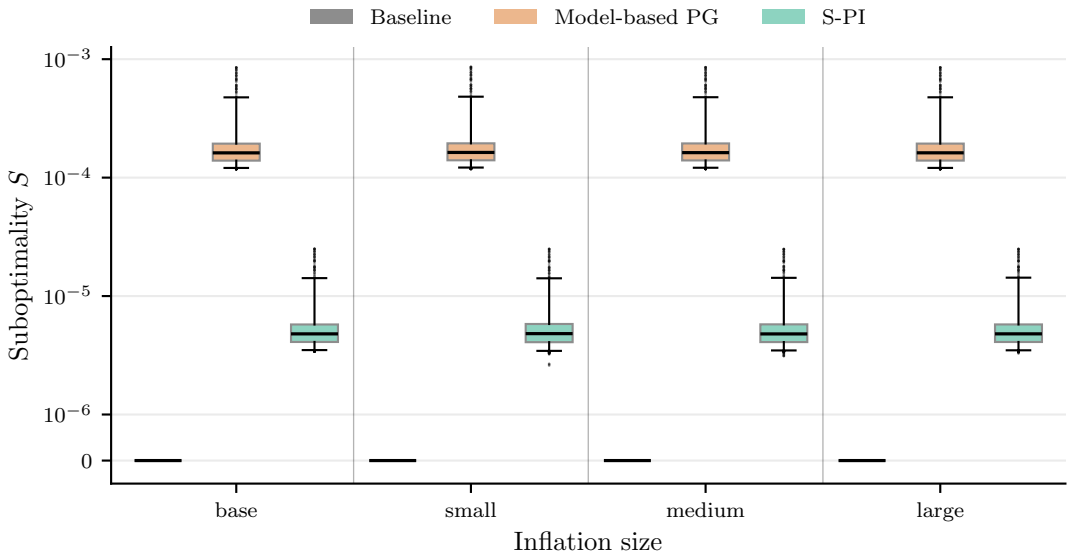


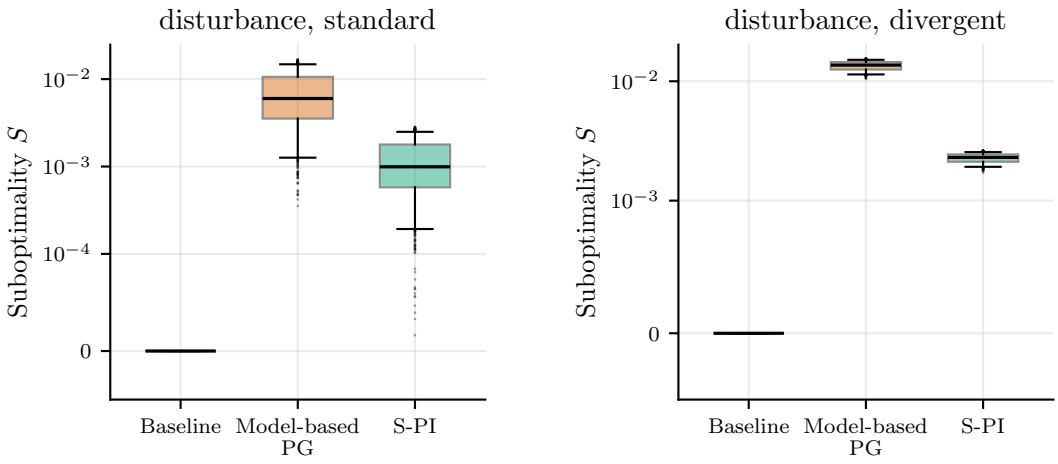
Figure 3.1: Suboptimality S of the solvers on the true model for each inflation size.

Overall, this looks very promising for S-PI; still, our goal is to find out whether it is also a robust controller.

3.1.3 Noise Experiments

Now, we reuse the controllers we trained in Section 3.1.2 and evaluate them under the test protocol with noise (Section 2.3.1.2) injected.

When looking at the results, one can notice that our baseline controller, which was optimal for the clean loop, mostly performs worse than our trained controllers, except for external disturbance. However, when only comparing Model-based PG with S-PI, we can see differences in their performance, depending on the type of noise that is applied, as well as whether the loop is inflated or not. For parametric uncertainty, no real difference can be seen between the mean performance of all controllers; however, Model-based PG shows a higher variance. When looking at external disturbance under *large* inflation, one can notice that with increasing noise it becomes clearer that S-PI is more robust here compared to Model-based PG. While for *weak* and *standard* their total spread still shows some overlap, for *strong* and *divergent* S-PI shows a cleanly separated suboptimality below Model-based PG (Fig. 3.2).



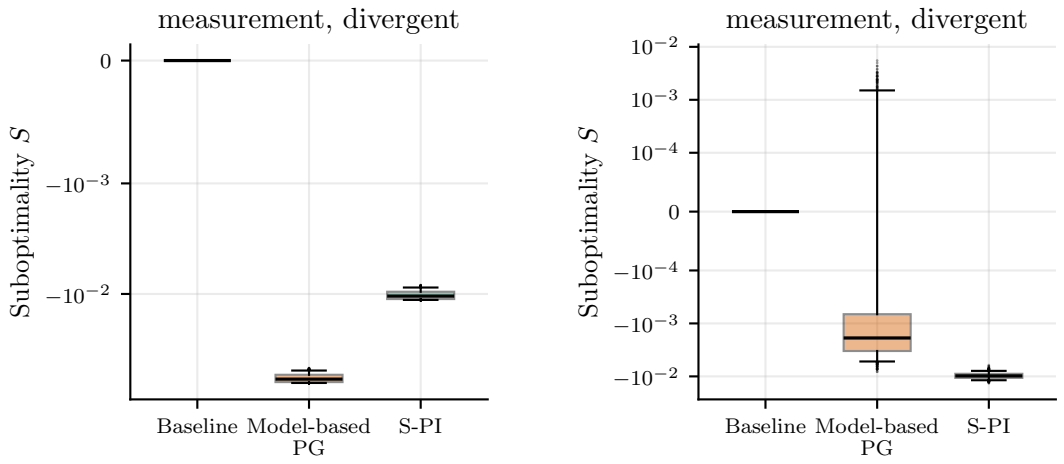
(a) Intensity *standard*.

(b) Intensity *divergent*.

Figure 3.2: Suboptimality S of the solvers on the true model under *large* inflation and external disturbance at different intensities.

For control noise in comparison, we can see that here Model-based PG is more robust on average across settings. However, its worst-case performance here is worse than that of S-PI, so it has an overall wider spread of suboptimalities. Still, both mostly outperform the baseline controller. Measurement noise becomes an interesting case here. While both controllers outperform the baseline, under no inflation, Model-based PG seems to be much more robust here. However, when looking at the controllers under *large* inflation with the same noise level applied, Model-based PG deteriorates here, with the spread now reaching suboptimality values worse than baseline. Compared to that, S-PI shows more robustness here

with little variation and stays at a similar value, better than baseline. Figure 3.3 visualizes this.



(a) *base* inflation.

(b) *large* inflation.

Figure 3.3: Suboptimality S of the solvers on the true model under measurement noise at intensity *divergent*.

Overall, for the *strong* level and *large* inflation, this is how both controllers compare to the baseline and each other across the noise channels (Table 3.2).

Table 3.2: Suboptimality S (in 10^{-3}) under *large* inflation and each noise channel at intensity *strong*.

Solver	none	measurement	control	parametric	disturbance	all channels
Model-based PG	$+0.2 \pm 0.12$	-1.5 ± 1.8	-12 ± 5.2	-1.6 ± 23	$+11 \pm 3$	-29 ± 48
S-PI	$+0.0059 \pm 0.0037$	-7.8 ± 2.2	-2.2 ± 0.9	-0.39 ± 4	$+1.8 \pm 0.5$	-9.1 ± 16

3.1.4 Discussion

Generally, we note that trained controllers are mostly at an advantage compared to statically calculated controllers with regard to robustness. By not achieving full optimality on the nominal problem, they can endure measurement and control noise better; however, they are still more affected by external disturbances. When comparing now trained fully parameterized and structured approaches, differences definitely appear. Changes in the underlying model, as in parametric uncertainty, affect all controllers mostly the same, with on average no real difference to the baseline. Looking at the other cases, we find that low-dimensional controllers actually offer better robustness in strong noise related to the inputs of the controller, especially the larger the input dimension. Measurement noise shows this clearly, as the performance of an overparameterized controller degrades here with increasing

size, while a regularized controller stays robust across sizes. This stems from the construction of our initial policy. As the full-rank component of K_0 is not affected by any inflated input on the nominal plant, Model-based PG, contrary to S-PI, does not remove it. These components then become costly under measurement noise, whereas S-PI stays resistant. However, related to the output dimensions, under control noise reduced controllers perform worse on average, but still better in the worst case than the full controller. And under *large* size the reduction in needed parameters is two orders of magnitude, which shows the efficiency of the reduced controller.

3.2 Unknown Model across Dimensions

Now we take a look at the more realistic case, where we do not know the underlying model, and compare different approaches. Before we can get into the results, we first take a deeper look at the solvers that we use and how the experiments are set up.

3.2.1 Setting

First, we introduce our basis for the indirect model-free approach, which is based on system identification. N4SID, introduced by Van Overschee and De Moor [46], uses "the projection of the future outputs [of the system] onto the past and future inputs and the past outputs" [46, Sec. 3]. These projections are then called "a function of the system matrices and the input-output block Hankel matrices" [46, Sec. 3], which allows us to reconstruct exactly that from the results. Important are further the assumptions that the systems are observable (Definition 2.4) and controllable (Definition 2.2). For simplicity, we rely on the implementation of N4SID from SIPPY [CITE: TODO], which we integrate into our own setup. This implementation allows us to recover the system matrices A, B and C, whereas the cost matrices Q and R are choices of the controller design (Section 2.3.3), and thus already known beforehand.

The collection of observation and action data runs the plant for T steps under the initial gain, with a randomly sampled excitation added on top of it,

$$v_k = -K_0 y_k + \sigma_e w_k, \qquad w_k \sim \mathcal{N}(0, I_{d_{\text{out}}}), \qquad k = 0, \dots, T-1. \tag{3.5}$$

The paper mentions that the approximation error becomes smaller as the number of block rows of the Hankel matrices grows [46, Sec. 5.3]. With the number of columns being dependent on our trajectory length T, we aim to find the best combination of both. Denoting the number of block rows as the "Hankel horizon", we evaluated different combinations to find the optimal setting to use in our experiment. The full results of the selection can be found in Section C.3. Specifically, we use the settings in Table 3.3.

Table 3.3: Identification settings per inflation size.

Name	Trajectory length T	Hankel horizon
base	500	5
small	500	5
medium	500	5
large	2000	5

The resulting identified system then serves as a surrogate for the true model for the model-based approaches from Section 3.1.1. We use for N4SID + Model-based PG an $\eta = 10^{-4}$ and for N4SID + S-PI $\lambda = 1$ across all sizes. This thus produces results without ever having access to the real model.

An attentive reader might now ask why we would even want to use these iterative approaches, if we could just solve the DARE (2.20) directly on this estimation. And indeed we include this N4SID + DARE controller as an indicator for how well N4SID identifies the system in our following comparisons.

Additionally, we also use direct model-free approaches, which are variations of the model-based approaches from Section 3.1. Unlike before, they do not calculate the gradient directly, but estimate it based on the returns of N_{traj} pairs of perturbed versions $K \pm \Xi_i$ of the current policy K, each simulated for l steps. As one might assume, more "data" could conversely lead to better estimation and thus better policy updates, but it requires more computation. So, to settle this, we evaluated different combinations of them in Section C.4. The best trade-off between computational efficiency and quality gave $N_{\text{traj}} = 200$ and $l = 500$.

We use the model-free natural policy gradient (NPG) estimation described by Fazel et al. [12], which at its core consists of the loop in Algorithm 3. Instead of the proposed single-point estimator, we use a two-point estimator inspired by Duchi et al. and Mania et al. [10, 24], as this drastically reduced the variance of the estimates, and thus the instability of the controller.

We further aim to clarify a couple of things about the given loop. Subtracting the mean difference $\bar{\delta}$ does not change the expected value of the estimate, since $\mathbb{E}[\Xi_i] = 0$, but lowers its variance. A rollout whose observation exceeds $\|y_k\| = 10^5$, we count as diverged. It is therefore ended early and charged a penalty cost of 10^{10}. Additionally, for inflated dimensions, $\hat{\Sigma}_K$ is singular, so we precondition with its truncated pseudoinverse $\hat{\Sigma}_K^{\dagger}$ instead, which discards singular values below 10^{-6} times the largest one.

Our full learning algorithm is then given in Algorithm 4.

After some evaluations, we have decided on a smoothing parameter of $r = 0.5$, and a learning rate of $\eta_{\text{base}} = 0.1$, $\eta_{\text{small}} = 0.03$, $\eta_{\text{medium}} = 0.01$ and $\eta_{\text{large}} = 0.01$ based on the results in Section C.5.

Algorithm 3 Two-point gradient estimation

Require: gain $K \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, number of perturbation pairs N_{traj}, rollout length l, smoothing radius r

for $i = 1, \dots, N_{\text{traj}}$ **do**
  sample Ξ_i uniformly from the matrices with $\|\Xi_i\|_F = r$
  sample an initial state $x_0 \sim \mathcal{D}$
  simulate $K + \Xi_i$ and $K - \Xi_i$ for l steps from x_0, with stage costs $c_k^{\pm}$ and observations $y_k^{\pm}$
  $\hat{J}_i^{\pm} = \sum_{k=0}^{l-1} c_k^{\pm}, \quad \hat{\Sigma}_i^{\pm} = \sum_{k=0}^{l-1} y_k^{\pm} \left(y_k^{\pm}\right)^{\top}, \quad \delta_i = \hat{J}_i^{+} - \hat{J}_i^{-}$
end for
$\bar{\delta} = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \delta_i$
$\widehat{\nabla f}(K) = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \frac{d_{\text{out}} d_{\text{in}}}{2r^2} \left(\delta_i - \bar{\delta}\right) \Xi_i$
$\hat{\Sigma}_K = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \left(\hat{\Sigma}_i^{+} + \hat{\Sigma}_i^{-}\right)$
$\hat{f}(K) = \frac{1}{N_{\text{traj}}} \sum_{i=1}^{N_{\text{traj}}} \frac{1}{2} \left(\hat{J}_i^{+} + \hat{J}_i^{-}\right)$
return $\widehat{\nabla f}(K)$, $\hat{\Sigma}_K$, $\hat{f}(K)$

Algorithm 4 Model-free NPG

Require: stabilizing gain $K_0 \in \mathcal{K}$, N_{traj}, l, r, step size η

for $j = 0, 1, 2, \dots$ **do**
  compute $\widehat{\nabla f}(K_j)$ and $\hat{\Sigma}_{K_j}$ by Algorithm 3
  update the gain by $K_{j+1} = K_j - \eta \, \widehat{\nabla f}(K_j) \, \hat{\Sigma}_{K_j}^{\dagger}$
end for

Next, we implement a model-free version of S-PI that relies on the same gradient estimation (Algorithm 3) as Model-free NPG. Another crucial difference, apart from the estimation of the gradient compared to the calculation, is the "usage of a fixed stepsize η, rather than an adaptive stepsize from a backtracking linesearch" [29, Sec. 3.1], as that would require access to the function value $f(K) = \operatorname{tr}(P_K \Sigma_0)$. Before we look at how we select this step size and λ, we define our learning loop in Algorithm 5.

Algorithm 5 S-PI (model-free)

Require: stabilizing gain $K_0 \in \mathcal{K}$, N_{traj}, l, r, fixed step size η, weight λ, tolerance ε

for $j = 0, 1, 2, \dots$ **do**
  compute $\widehat{\nabla f}(K_j)$ by Algorithm 3
  compute K_{j+1} by (3.3), with $\widehat{\nabla f}(K_j)$ in place of $\nabla f(K_j)$
  if $\|K_{j+1} - K_j\|_F \leq \varepsilon$ **then**
    stop
  end if
end for

Now to the selection of the parameters. Park et al. already note the difficulty of finding a fixed step size that does not produce divergence of the method [29]. Also, the example shown only uses the model-based approach and the regularizer to induce sparse structure, not the model-free variant combined with low dimensionality. So during our first tests, we often ran into either unstable controllers, or ones that did not induce any structure at all, hinting at a too low λ. If we remember the proximal step (3.3), we can see that it depends on both the step size and λ, as every singular value is mapped to

$$\sigma_i \mapsto \max\{\sigma_i - \eta\lambda, 0\}. \tag{3.6}$$

Even after some improvement by using the two-point estimator (Algorithm 3), it remained quite a difficult topic. So rather extensive experiments on the variation of λ and the step size can be found in Section C.6. Based on those, we selected $\lambda_{\text{base}} = 4$ and $\eta_{\text{base}} = 3.5 \times 10^{-6}$, $\lambda_{\text{small}} = 400$ and $\eta_{\text{small}} = 3.5 \times 10^{-5}$, $\lambda_{\text{medium}} = 400$ and $\eta_{\text{medium}} = 3.5 \times 10^{-5}$; and $\lambda_{\text{large}} = 40\,000$ and $\eta_{\text{large}} = 3.5 \times 10^{-8}$.

3.2.2 Results

Looking at the time until convergence, as well as the achieved training suboptimality, we notice that S-PI (model-free) produced under *large* inflation a very expensive gain, while still not achieving a strong reduction in rank. As this was the same problem we observed during our parameter search (Section C.6), we accept it as an unsuccessful attempt and difficult to implement. Apart from this, it is noticeable that Model-free NPG takes longer to converge the larger the inflation gets, with it being much slower than N4SID + S-PI and N4SID + Model-based PG, where N4SID + S-PI again is still much faster than N4SID + Model-based PG. The fastest approach here is the DARE solution, as it only requires the identification step, and no iteration.

Looking at the policy sizes, we can see that N4SID + S-PI indeed reduces its size to that of the optimal policy, with the DARE policy also being smaller than the iterative versions, but still slightly larger (Fig. 3.4).

Analyzing the resulting test suboptimality, we see the expected worse performance of S-PI (model-free), but also that the direct approaches underperform the indirect (identification) approaches. Here, we can also notice a difference: with increasing inflation size, the Model-free NPG trains for more iterations, with an increase from 167 at *base* to 610 at *large*, while for the identification-based policies it stays similar. What also jumps out is that N4SID + DARE in some cases even achieves optimal performance (Fig. 3.5).

3.2.3 Noise Experiments on Large Inflation

We first start by analyzing the effects we can observe on the uninflated system under noise. For parametric uncertainty, the spread of suboptimality again ranges

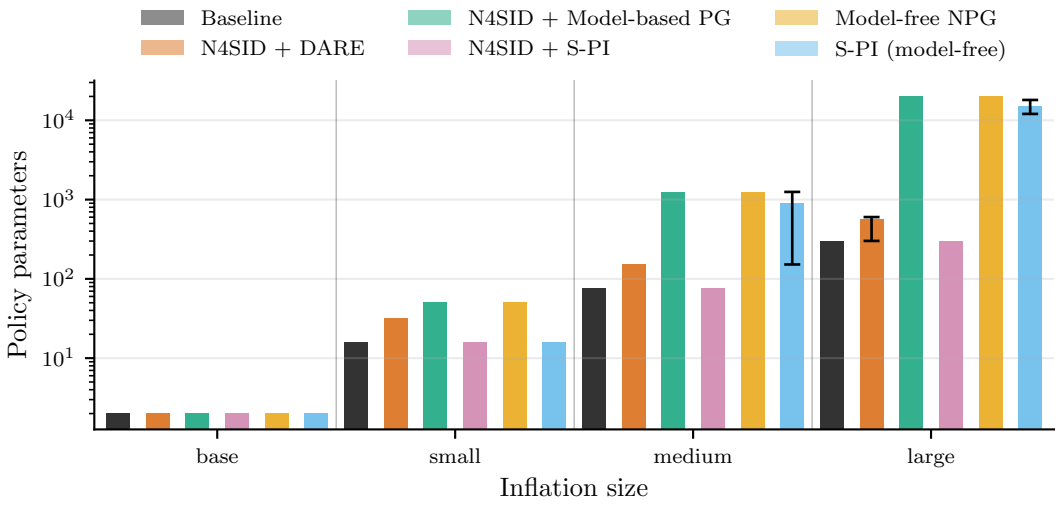


Figure 3.4: Policy size $|K|$ (Definition 2.6) learned without the model, for each inflation size.

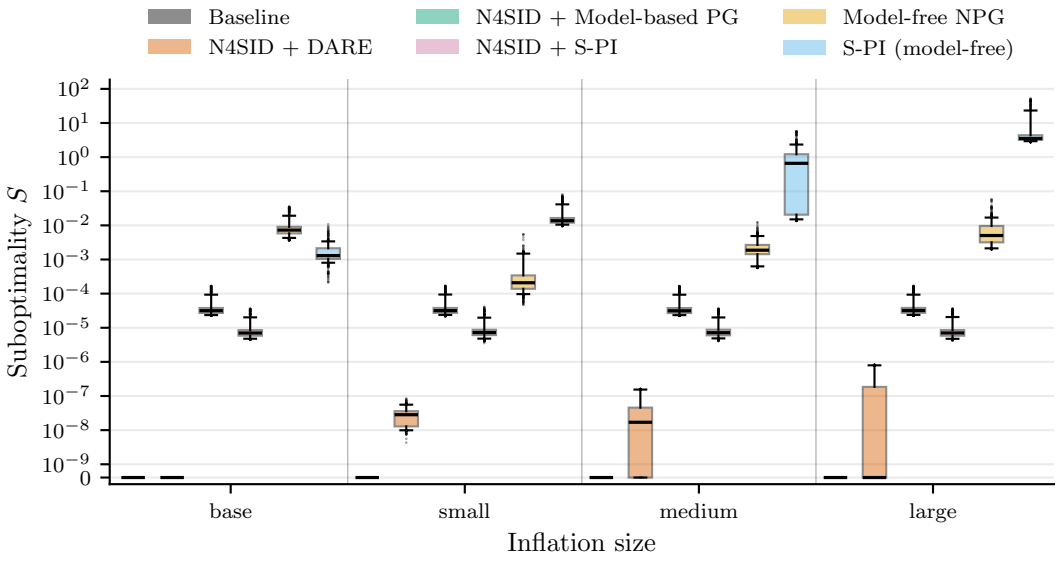
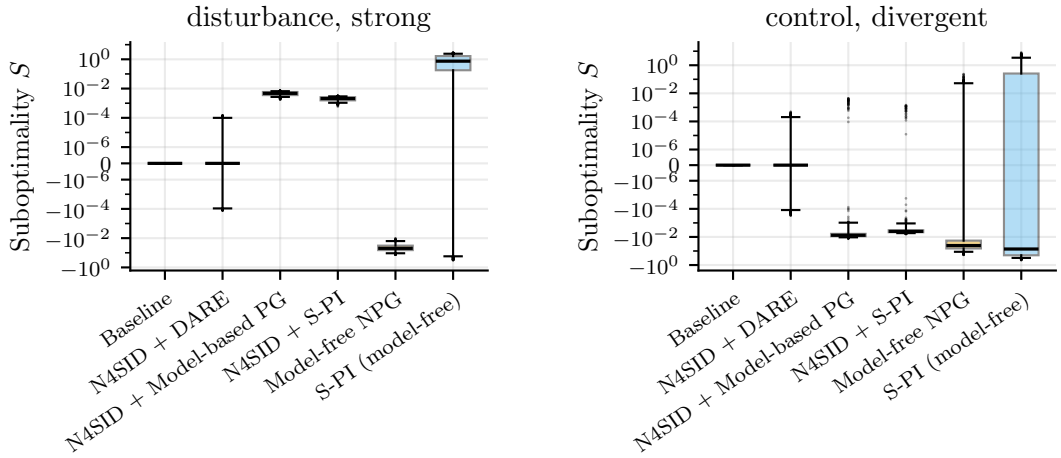


Figure 3.5: Suboptimality S of the solvers without the model for each inflation size.

from $+0.1$ to -0.1 for all the iteratively trained approaches, with the DARE solution having a much smaller spread, centered around the baseline. Still, it is observable that the N4SID-based solvers perform on average much better than the direct approaches. For control noise a similar structure can be noticed; here, however, the spreads separate much more clearly into worse for direct approaches and better for indirect approaches. For both external disturbance and measurement noise, the DARE solution remains closely spread around 0. For external disturbance, all the other solvers except for Model-free NPG perform worse than the

baseline, while Model-free NPG sits at a suboptimality of -0.1 for *divergent*. For measurement noise, the image suddenly flips, with Model-free NPG performing worse, while all the other approaches sit at a suboptimality of around -0.01.

Now inflating the system to *large* size and repeating our experiments, we see similar observations for parametric uncertainty, except for S-PI (model-free), which performs worse than the others by multiple orders of magnitude. Looking at control noise, on average Model-free NPG performs best here, even though the spread under *strong* or *divergent* control noise admits episodes worse than the baseline. Iterative N4SID solvers in comparison here do also outperform the baseline, but are slightly worse than Model-free NPG. However, they remain consistently below baseline across all intensities and runs. The most interesting observation here is that, even though S-PI (model-free) performs much worse on the clean plant, its median is on par with Model-free NPG under *divergent* control noise (Fig. 3.6), even though its mean lies above the baseline.



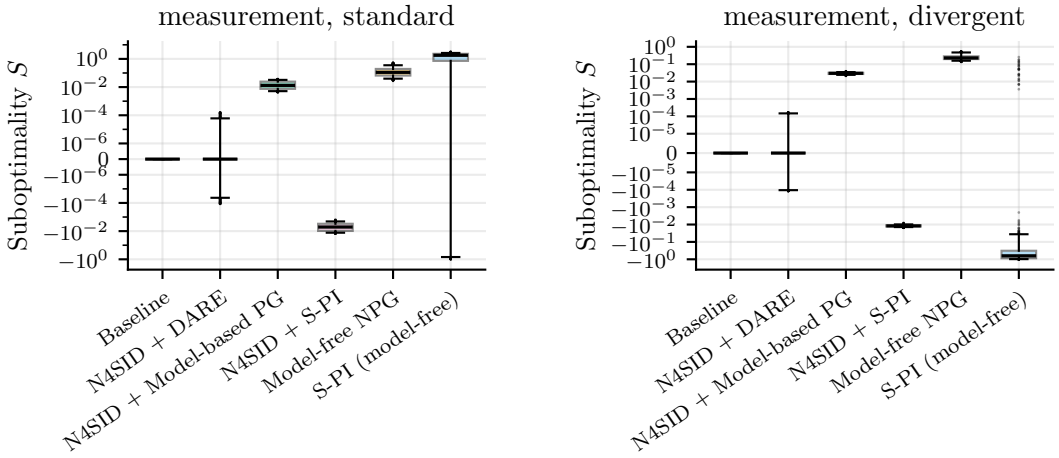
(a) External disturbance at intensity *strong*.

(b) Control noise at intensity *divergent*.

Figure 3.6: Suboptimality S of the solvers without the model under *large* inflation.

For external disturbance, the indirect solvers perform worse here, while Model-free NPG is clearly the robust choice here, under any strength. Again, it is noticeable that the stronger the noise gets, the more runs of S-PI (model-free) achieve a negative suboptimality. While on average still worse, under *divergent* intensity, some do indeed perform the same as Model-free NPG. This phenomenon can also be observed with measurement noise: here, S-PI (model-free) is on average at a suboptimality of 10, 1 and 0.1 for *weak*, *standard* and *strong*, but for *divergent* it then clearly sits at -1. With increasing intensity, the spread can be seen shifting more and more downwards. Between indirect solvers, N4SID + S-PI is the only one that stays below baseline at every intensity, performing better than N4SID + Model-based PG, which differs from it only by its full-rank gain. Overall, it can

be seen that the reduction-based solvers do perform better here than their fully parameterized counterparts under *divergent* intensity (Fig. 3.7).



(a) Intensity *standard*.

(b) Intensity *divergent*.

Figure 3.7: Suboptimality S of the solvers without the model under *large* inflation and measurement noise.

3.2.4 Discussion

Overall, we notice that for a clean plant, identifying the system and solving the DARE based on these observations is the fastest solution, and even under noise it achieves a performance similar to the baseline solution. A good alternative is offered by approaches that use the same identified system and combine it with model-based approaches. They offer, in the case of S-PI, a much shorter training than direct approaches, and, due to their suboptimality on the nominal loop, they are more robust to control noise. Compared to them, direct approaches require much more training, and perform worse on the clean system. The only exception is external disturbance, where Model-free NPG, as a fully parameterized, iteratively trained policy, outperforms all other approaches.

This distinction between fully parameterized and dimensionally reduced must also be made. While the direct S-PI (model-free) approach varies quite a lot in its performance, under measurement noise on inflated systems, it improves in comparison to its counterpart, and even outperforms it under *divergent* noise. N4SID + S-PI compared with N4SID + Model-based PG does so across all measurement noise intensities.

Combining both of these distinctions, in most cases, except for external disturbance, an N4SID + S-PI-based controller is the best choice. It offers the advantages of fast training of N4SID, better performance than baseline under control noise, together with the robustness under measurement noise from the reduced

dimensionality, while also being orders of magnitude smaller. Solving the DARE directly also offers a good solution, being stable across all experiments, but it is often outperformed by the reduced controller.

3.3 Discussion

We take from this chapter that direct approaches are mostly outperformed by indirect approaches, which try to identify the model first. They require around four orders of magnitude of training steps less, and achieve performances close to optimal on the clean plant. By not reaching full optimality, they are less affected by control noise than the DARE solution on the clean plant. Regarding noise, it is also noticeable that the strength of low-dimensional controllers is their resistance to measurement noise, which becomes much clearer under increasingly larger dimensions as full-rank parameterizations do not remove components in the null space. With these learnings in mind, we will now progress to our RL approaches next.

Chapter 4

Robustness of Reinforcement Learning Approaches

Having seen how low-dimensionality affects robustness with a simple, LQR-based controller, we now want to take a look at a setting which allows us to judge architectures used in more complex tasks as well. Still, we will stay with our introduced double integrator system, but change the controllers for now.

4.1 PPO and Its Variants

4.1.1 Proximal Policy Optimization

Before getting into our implementation details, we want to explain the basic construction of PPO. Introduced by Schulman et al. [38] in 2017, it is an actor-critic algorithm, which means it consists of an actor, guided by a policy, that is simulated in an environment, and a critic, which tries to estimate the underlying value function based on the current policy and the actor's returns [20]. In the case of PPO, as we are trying to optimize a policy via policy gradient optimization, we "[construct] an objective function whose gradient is the policy gradient estimator" [38, Sec. 2.1]. What makes PPO special is the construction of this surrogate function, based on ideas from Trust Region Policy Optimization [37]. In the paper, two approaches are proposed: a Clipped Surrogate Objective or an Adaptive Kullback–Leibler (KL) Penalty Coefficient. The former clips the probability ratio past a limit set by a hyperparameter, imposing constraints on the policy update size [38, Sec. 3], while the latter, rather than setting a fixed KL for an update, sets a target KL divergence [38, Sec. 4]. In case further interest arises, we refer to the original paper [38]. With the probability ratio $r_k(\theta)$ between the current policy π_θ

and the policy $\pi_{\theta_{\text{old}}}$ that collected the data, the clipped objective becomes

$$\begin{aligned} r_k(\theta) &= \frac{\pi_\theta(v_k \mid y_k)}{\pi_{\theta_{\text{old}}}(v_k \mid y_k)}, \\ L^{\text{CLIP}}(\theta) &= \hat{\mathbb{E}}_k \left[\min \left\{ r_k(\theta)\hat{A}_k,\ \text{clip}\left(r_k(\theta), 1-\varepsilon_{\text{clip}}, 1+\varepsilon_{\text{clip}}\right)\hat{A}_k \right\} \right]. \end{aligned} \tag{4.1}$$

4.1.2 Training Setup

However, as has been noted before [11, 14], the actual implementation of PPO matters, and can influence the effectiveness of it in practice. For our construction, we use the architecture provided by Stable-Baselines3 (SB3) [40]. As the standard architecture, for our Fully-connected PPO, we use the standard two-layer multi-layer perceptron (MLP) with 64 neurons for both the actor and the critic [39]. Additionally, unless named otherwise, we will use the standard parameters from the SB3 library. For the above named parts, we leave the clip range at the standard of 0.2 and set the target KL to 0.02. Experimenting with different runtime lengths, we settled on the following values, which differ by inflation size:

Across all, the evaluation frequency, so when the policy of the actor is updated, is set to 25 000.

After initial runs with this basic setup, we could observe some divergence behavior, which led us to consider additional implementation improvements, as suggested by Huang et al. [15]. Specifically, we started using a vectorized environment, similar to the paper [38]. Additionally, we modified the discount factor to $\gamma = 0.999$ instead of 0.99. However, the two most effective changes were:

1. Using learning rate annealing instead of a fixed rate [15, Detail 4]. Now the learning rate is decreased with every update, such that at the end of the number of steps (Table 2.3) it is equal to 0. So this is

$$\eta(N_{\text{env}}) = \eta_0 \max\left\{1 - \frac{N_{\text{env}}}{N_{\max}},\ 0\right\}. \tag{4.2}$$

2. Separating the feature extractors for actor and critic [15, Detail 13]. As described by [39], to reduce computation, in the standard implementation, they share the extractor, which is "a simple MLP network consisting of two layers of 64 neurons and Hyperbolic Tangent as the activation function" [15]. Later, when introducing pretrained extractors, they will be shared again, but for still learning ones, we will separate them.

Algorithm 6 summarizes the resulting training loop.

4.1.3 Policy Architectures

Now, after having introduced the base construction, it makes it easier for us to describe variations of the same. We start with Linear PPO, which allows us to

Algorithm 6 PPO implemented here

Require: initial actor π_θ and critic V_ϕ, initial learning rate η_0, step cap $N_{\max}$, convergence parameters N_{w}, N_{p}, $\varepsilon_{\mathrm{rel}}$

$N_{\mathrm{env}} \leftarrow 0$
while $N_{\mathrm{env}} < N_{\max}$ **do**
  run π_θ for 2048 steps, sampling $v_k \sim \pi_\theta(\cdot \mid y_k)$; store y_k, v_k and the rewards
  normalize the rewards by a running estimate of the return's standard deviation
  compute the advantages $\hat{A}_k$ and value targets with V_ϕ
  $\theta_{\mathrm{old}} \leftarrow \theta$, set the learning rate $\eta(N_{\mathrm{env}})$ by (4.2)
  for epoch $= 1, \ldots, 10$ **do**
    for each minibatch of 64 stored steps **do**
      take an Adam step on $-L^{\mathrm{CLIP}}(\theta) + 0.5 \cdot$ (value loss of V_ϕ), with the gradient norm clipped to 0.5
    end for
    if the approximate KL divergence to $\pi_{\theta_{\mathrm{old}}}$ exceeds $1.5 \cdot 0.02$ **then**
      break
    end if
  end for
  $N_{\mathrm{env}} \leftarrow N_{\mathrm{env}} + 2048$
  if 25 000 steps have passed since the last check **then**
    evaluate π_θ, update g_j with the mean training cost since the last check
    if Definition 2.7 holds **then**
      stop
    end if
  end if
end while
return the policy with the best smoothed signal $\bar{g}_j$

compare the performance of PPO to the previous chapter. Here, our actor becomes a single layer, with the bias removed, such that we can rewrite this network as

$$\mu(y_k) = W y_k = -K y_k, \qquad K \coloneqq -W \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}, \tag{4.3}$$

which allows us to extract a gain matrix K that is comparable to the previous chapter. A second, already known, component is used for State encoder + PPO. Here, we reuse an identified system from N4SID (Section 3.2) to encode an observation into a latent state:

$$\hat{x}_k = \hat{C}^\top y_k, \qquad \hat{C} \in \mathbb{R}^{d_{\mathrm{in}} \times \hat{n}}, \quad \hat{C}^\top \hat{C} = I_{\hat{n}}. \tag{4.4}$$

The input dimension for both the actor and the critic thus becomes $\hat{n}$, which is the order of the identified model.

However, as we will later scale our inputs to larger dimensions (Chapter 5), N4SID will become infeasible for such dimensions. Still, we want to stay in line with the

idea of encoding our observations into a latent state, as this exactly introduces the bow-tie structure we want. So, we introduce another encoder, which offers this scalability, and call the combination Scalable encoder + PPO. This feature extractor is based on the results from [17, 3], which use robotic priors. Robotic priors, simply described, are prior knowledge, like physical laws, from which we define loss functions. By adding them to the training process, and optimizing for them, we can ensure that the learned representation is in line with these laws. As mentioned in Section 2.4, this allows us to make use of the embodiment to improve our controller architecture. Here, we will use the following priors [3]:

- simplicity: the state is lower in dimensionality than the observation,
- temporal coherence: state changes are rather slow,
- proportionality: same actions should show similar output strength,
- repeatability: same actions should show similar output changes,
- causality: TODO

We modify the loss calculations taken from the paper [3] slightly, by dropping the Lreg term, and thus get the total loss function of

$$\mathcal{L} = \omega_1 \mathcal{L}_{\text{temp}} + \omega_2 \mathcal{L}_{\text{prop}} + \omega_3 \mathcal{L}_{\text{rep}} + \omega_4 \mathcal{L}_{\text{caus}}. \tag{4.5}$$

As this encoder must be trained first, we pretrain it for a total of 20 000 steps and 20 epochs (so 1000 steps per epoch), before freezing its parameters and prepending it to the actor and the critic. An important parameter is also the dimension of the latent state, which we set to the actual "2" here. Although for later application, one might not know the real, underlying dimension of the dynamics, we want to make it comparable across the solvers by using the same low dimension. And as N4SID mostly correctly identifies the true dimension, we use it here as well. Algorithm 7 summarizes the pretraining.

The second trainable extractor we used is based on Todorov et al.'s paper [45] using orthogonal bottlenecks. It consists of a fully connected MLP, with 512 neurons in the first, and 256 neurons in the second layer. Following it is a matrix with orthonormal columns, which projects the output into a state of bottleneck dimension d_{b}, which we here set to 2. Like U_1 in (2.26), the matrix is the orthonormal factor of the QR decomposition of a random matrix with standard normal entries, so that the feature extractor output $z_k \in \mathbb{R}^{d_{\text{z}}}$ is mapped to

$$U_{\text{b}}^{\top} z_k \in \mathbb{R}^{d_{\text{b}}}, \qquad U_{\text{b}} \in \mathbb{R}^{d_{\text{z}} \times d_{\text{b}}}, \quad U_{\text{b}}^{\top} U_{\text{b}} = I_{d_{\text{b}}}. \tag{4.6}$$

Different to the previous encoder, this one is trained together with PPO, so we just combine them.

The following figure visualizes the architecture for all our solvers.[CITE: figure below]

Algorithm 7 Pretraining of the scalable encoder

Require: initial gain K_0, action bound $a_{\max}$, encoder $f_\psi \colon \mathbb{R}^{d_{\text{in}}} \to \mathbb{R}^2$, weights $\omega_1, \ldots, \omega_4$

for $k = 0, \ldots, 20\,000 - 1$ **do**
  with probability 0.3, draw v_k uniformly from $[-a_{\max}/10, a_{\max}/10]^{d_{\text{out}}}$, otherwise set $v_k = -K_0 y_k$
  apply v_k, store (y_k, v_k, y_{k+1}), and reset the plant after 800 steps
end for
for epoch $= 1, \ldots, 20$ **do**
  for each minibatch of 128 stored transitions **do**
    pair every transition with a randomly drawn second transition
    take an Adam step on $\omega_1 \mathcal{L}_{\text{temp}} + \omega_2 \mathcal{L}_{\text{prop}} + \omega_3 \mathcal{L}_{\text{rep}} + \omega_4 \mathcal{L}_{\text{caus}}$ of these pairs
  end for
end for
standardize the latent state with its mean and standard deviation over the stored observations
freeze ψ and place f_ψ in front of the actor and the critic

4.2 Comparison across Dimensions

Now that we have all our methods set up, we can train and compare them across the different inflation sizes.

4.3 Noise on Large Inflation

4.4 Discussion

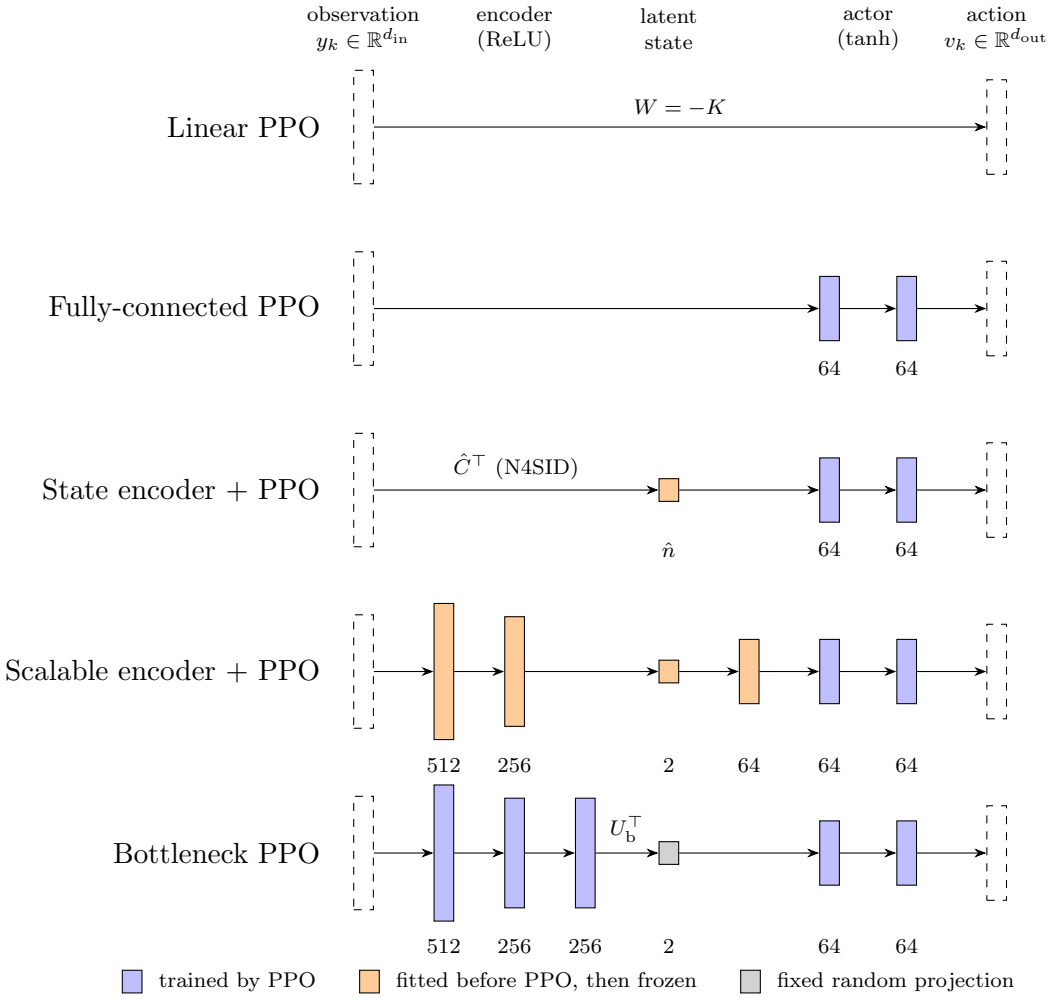


Figure 4.1: Actor architectures of our PPO variants. For each layer, the number below indicates its neurons. The critic receives the same encoder structure and uses its own 64-64 tanh perceptron with a scalar output. Visualization created with Claude Code.

Chapter 5

Perception-Based Experiments

From the previous artificially inflated inputs, we aim to move to a more realistic example, which shows the robustness of low-dimensional controllers in a scenario involving an image observation as input. We do this by first using our setup from the previous chapter to scale it to a dimension, which resembles image inputs. Afterwards, we actually build a new environment, which combines perception-based observation with multiple actuators acting on our double integrator.

5.1 Artificial Inflation to Pixel Size

5.1.1 Setting

As mentioned, we take our loop from before (Section 2.3.1.1) and scale it up further. The input dimension in this case becomes 18 432, while the output dimension stays at 256. These numbers were specifically selected to match the setup in Section 5.2, such that we can compare both more closely. For the input dimension, this is a 96×96 image stack of two images, while for the output dimension it was chosen due to the specific construction.

We further note that State encoder + PPO cannot be applied at this scale, as the identification for such large matrices is computationally impossible. Additionally, we excluded Linear PPO, as it was only used to compare with LQR approaches.

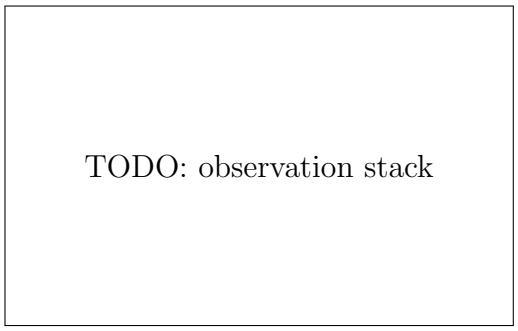
5.1.2 Results

5.2 The New Environment and the Position-Velocity Encoder

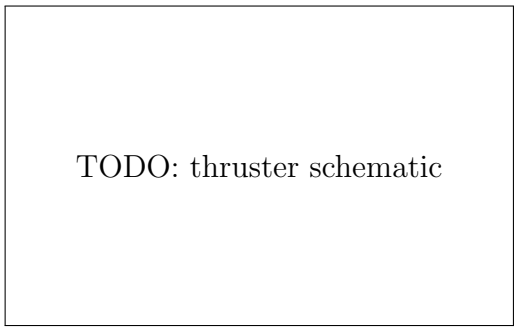
5.2.1 Pixelated Environment

For our perception setup, we still use the underlying double integrator as a basis, however now simulate its actual movement. This simulation is observed by a monochromatic camera producing 96×96-pixel images of a mass with radius of z pixels moving along a horizontal line. To be able to see changes between the pictures, we hold any action commanded by the agent for 30 time steps before capturing another frame. The agent then receives a stack of two images as the input. Figure 5.1a shows an example of such an image.

But not only the input is high-dimensional compared to the actual state. We also modify the way our mass is actuated. We also multiply the number of actuators to 256, distributed equally around a half circle, with their thrust direction acting on different angles on the mass. Figure 5.1b visualizes the thrust directions for eight thrusters.



(a) TODO



(b) TODO

Figure 5.1: TODO

We deliberately chose the action repeat and scaling of the frame, as well as other parameters such that it is possible to observe position changes, and thus possibly velocity, from two different frames. For a better approximation the number of frames stacked could be increased, or anti-aliasing could be used to display changes better. But this will further be discussed in Section 5.5.

5.2.2 Position-Velocity Encoder

For this new environment, we introduce another feature extractor, which is based on robotic priors used in Scalable encoder + PPO [17]. Called Position-Velocity Encoders [18], it tries to encode a position and velocity state from an observation, by learning a latent position state, and then reconstructing the velocity state using

temporal differences. As we specifically look for position and velocity in our states, we can construct even more specific priors, again in line with our reasoning that knowledge about the embodiment helps in designing the controller architecture (Section 2.4). The priors used in [18, Sec. III-B] are:

1. Variation: "Positions of relevant things vary."
2. Slowness: "Positions change slowly."
3. Inertia: "Velocities change slowly."
4. Conservation: "Velocity magnitudes change slowly."
5. Controllability: "Controllable things are relevant."

For our implementation, we follow the paper's recommended approach. Each frame is input into a convolutional neural network (CNN), which encodes a latent position state $s_k^{(\mathrm{p})}$ from it. The dimension of it is a hyperparameter, but recommended to be set to five [18, Sec. III-C], which we follow. The velocity is then calculated as the difference between two frames, or $s_k^{(\mathrm{v})} = \alpha \left(s_k^{(\mathrm{p})} - s_{k-1}^{(\mathrm{p})} \right)$, where α is the inverse of the time between two agent actions. For the weighting of our prior loss terms, we use the combination suggested for similar tasks in [18, Tab. I], and thus exclude the controllability prior. For the training process we further follow the one proposed in [18, Sec. III-C]. For details on how exactly the total loss function is constructed, we refer the reader to the original paper [18]. Similar to the previous chapter, we train this extractor for a total of 20 000 steps, here split into 1 000 episodes of 20 steps, before freezing its weights and using it in front of our PPO construct (Section 4.1).

Scalable encoder + PPO is pretrained with a similar data split for this setting.

5.3 Effect of Visual Inputs on the Solvers

5.3.1 Setting

For the experiments here, we again compare the performance of the optimal controller K^* with our different approaches, which here consist of Pixels + PPO, Scalable encoder + PPO, Bottleneck PPO as well as our Position-Velocity Encoder + PPO. The optimal controller here only gives us the cost to compare it to, as it actually never acts based on the perception and with the thrusters, but instead on the true loop. For the other approaches, except for Position-Velocity Encoder + PPO, which takes images as an input, we flatten the images, such that they receive an 18 432-dimensional input vector.

5.3.2 Results

5.4 Noisy Perception

5.4.1 Setting

As our environment setup changed drastically, the cost increases associated with the noise levels calculated from before no longer apply. Thus, we again ran a calibration with the optimal controller to set them to respective levels (Section A.3). With these new values, we can use them like before to change how the loop behaves. Parametric uncertainty still changes the underlying model, external disturbance still acts on the position of our mass, and control noise changes the commanded actuation outputs. Measurement noise now affects the visual input, as shown in Fig. 5.2, by applying Gaussian noise to all the pixels in the image. Here, we set the levels ourselves, as the optimal controller is not acting from the camera image, but rather from the true state, so this noise does not affect it. Still, the chosen values are found in the appendix (Table A.3).

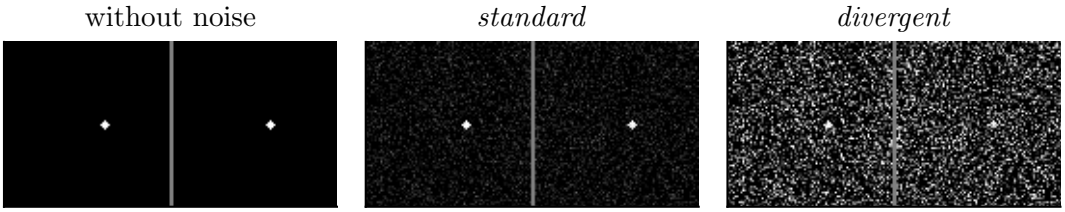


Figure 5.2: An observation without measurement noise, and with measurement noise of intensity *standard* and *divergent*.

We now run the trained controllers from Section 5.3 on noise-affected trajectories under different strengths according to our test protocol. To make the comparison under each noise channel equal, we exclude our optimal controller baseline, as it does not observe the measurement noise like the other controllers do. So instead we calculate the suboptimality for each controller in comparison to its clean run from Section 5.3.

5.4.2 Results

5.5 Discussion

Chapter 6

Conclusion

You generally cover three things in the Conclusions section, and each of these usually merits a separate subsection:

1. Conclusions
2. Summary of Contributions
3. Future Research

Conclusions are not a rambling summary of the thesis: they are short, concise statements of the inferences that you have made because of your work. It helps to organize these as short numbered paragraphs, ordered from most to least important. All conclusions should be directly related to the research question stated in the introduction. Examples:

1. The problem stated in the introduction has been solved: as shown in Sections ? to ??, an algorithm capable of handling large-scale Zylon problems in reasonable time has been developed.
2. The principal mechanism needed in the improved Zylon algorithm is the Grooty mechanism.
3. Etc.

The Summary of Contributions will be much sought and carefully read by the examiners. Here you list the contributions of new knowledge that your thesis makes. Of course, the thesis itself must substantiate any claims made here. There is often some overlap with the Conclusions, but that's okay. Concise numbered paragraphs are again best. Organize from most to least important. Examples:

1. Developed a much quicker algorithm for large-scale Zylon problems.
2. Demonstrated the first use of the Grooty mechanism for Zylon calculations.
3. Etc.

The Future Research subsection is included so that researchers picking up this work in future have the benefit of the ideas that you generated while you were working on the project. Again, concise numbered paragraphs are usually best.

Appendix A

Nomenclature

What goes in the appendices? Any material which impedes the smooth development of your presentation, but which is important to justify the results of a thesis. Generally it is material that is of too nitty-gritty a level of detail for inclusion in the main body of the thesis, but which should be available for perusal by the examiners to convince them sufficiently. Examples include program listings, immense tables of data, lengthy mathematical proofs or derivations, etc.

A.1 Notation

The following table contains the conventions regarding symbols used throughout this thesis, together with the place where each is defined:

Symbol	**Meaning**	**Defined in**
Sets and operators		
$\mathbb{R}$, $\mathbb{C}$, $\mathbb{N}_0$	real numbers, complex numbers, natural numbers including zero	
I_n, $\mathbf{1}_n$, e_1	identity matrix and all-ones vector of dimension n, first unit vector	
$M^\top$, $M^\dagger$	transpose and (truncated) pseudoinverse of M	Algorithm 4
tr, rank, diag	trace, rank, diagonal matrix	
$\lambda_i(M)$, $\sigma_i(M)$	i-th eigenvalue and i-th singular value of M	
$U\Sigma V^\top$	singular value decomposition, $\Sigma = \operatorname{diag}(\sigma_1, \ldots, \sigma_r)$	(2.40)
$\rho(M)$	spectral radius, $\max_i \lvert\lambda_i(M)\rvert$	Theorem 2.1

Symbol	Meaning	Defined in
$M \succeq 0$, $M \succ 0$	M symmetric positive semi-definite / positive definite	
$\|\cdot\|$, $\|\cdot\|_F$, $\|\cdot\|_*$	Euclidean (spectral) norm, Frobenius norm, nuclear norm	
$\|\cdot\|$	absolute value, cardinality of a set, size of a policy	Definition 2.6
$\odot$	elementwise product	(2.31)
$\mathbb{E}$, $\hat{\mathbb{E}}_k$	expectation, empirical average over the collected steps	(4.1)
$\mathcal{N}(\mu, \Sigma)$	Gaussian distribution with mean μ and covariance Σ	
∇f, $\widehat{\nabla f}$	gradient of f and its estimate	(3.2), Algorithm 3
System description		
n, m, q	dimension of the state, input and output	(2.1)
t, k, h	continuous time, discrete step index, sampling period	(2.3)
$x_k \in \mathbb{R}^n$	state	(2.4)
$u_k \in \mathbb{R}^m$	control input of the plant	(2.4)
$y_k \in \mathbb{R}^q$	observed output	(2.1)
p, v	position and velocity of the double integrator, $x = [\, p,\ v \,]^\top$	(2.7)
A_c, B_c	continuous-time system and input matrix	(2.1)
A, B	their zero-order-hold discretizations	(2.5)
C, D	output and feedthrough matrix; $D = 0$ throughout	(2.1)
A_K	closed-loop matrix, $A_K = A - BK$	(2.23)
$\mathcal{C}(A, B)$, $\mathcal{O}(C, A)$	controllability and observability matrix	(2.11), (2.14)
$\hat{x}_k$, $\tilde{x}_k$	state estimate and estimation error $x_k - \hat{x}_k$	(2.15)
L	observer gain	Definition 2.5
Optimal control		
Q, R	state and input cost matrices, $Q \succeq 0$, $R \succ 0$	(2.17)
$V_T(x_0, u)$	continuous-time cost over the finite horizon T	(2.17)
$J(x_0, u)$, $J(x_0, K)$	infinite-horizon cost of an input sequence or of a gain	(2.18)

Symbol	**Meaning**	**Defined in**
$K \in \mathbb{R}^{m \times n}$	feedback gain, $u_k = -Kx_k$	(2.19)
K^*, u_k^*	optimal gain and optimal input	(2.21)
P, $J^*(x)$	stabilizing solution of the DARE and optimal cost-to-go $x^\top Px$	(2.20), (2.22)
P_K	solution of the Lyapunov equation for a fixed K	(2.23)
$\mathcal{K}$	set of stabilizing gains	(2.24)
K_0	stabilizing initial gain, $K_0 = [\,0.1,\ 0.5\,]$	Section 2.3.3
$\mathcal{D}$, Σ_0	distribution of x_0 and its second moment $\mathbb{E}[x_0 x_0^\top]$	(2.25)
$f(K)$	expected cost of a gain, $\mathrm{tr}(P_K \Sigma_0)$	(2.25)
Inflated system		
d_{in}, d_{out}	inflated dimension of the controller input and output	(2.26)
U_1, U_2	lifts with orthonormal columns and rows	(2.26)
$K \in \mathbb{R}^{d_{\mathrm{out}} \times d_{\mathrm{in}}}$	inflated feedback gain, $v_k = -Ky_k$	(2.27)
v_k, $\tilde{v}_k \in \mathbb{R}^{d_{\mathrm{out}}}$	controller output, before and after control noise	(2.27), (2.31)
Q_n	state cost of the uninflated problem, $Q_n = U_1^\top Q U_1 = I_n$	(2.28)
Noise		
n_k^{m}, σ_{m}	measurement noise and its standard deviation	(2.30)
n_k^{c}, σ_{c}	control noise (relative) and its standard deviation	(2.31)
Δ_k^A, Δ_k^B, σ_{p}	parametric perturbations and their standard deviation	(2.33)
n_k^{d}, σ_{d}	external disturbance and its amplitude	(2.34)
c, $\varnothing$	noise condition (channels and intensity), noise-free condition	(2.37)
ℓ, c_ℓ, κ_ℓ	noise intensity, a channel at intensity ℓ, its cost ratio	(2.41)
Evaluation and training protocol		
T	horizon of a rollout	(2.37)
s, i	index of the seed and of the episode	(2.37)
M, N	number of seeds and of episodes per seed	(2.42)

Symbol	Meaning	Defined in
$J_T^{(i,s)}(K;c)$	empirical cost of one episode	(2.37)
$\hat{J}_T(K;c)$, $\mathcal{F}$	mean empirical cost, set of finite episodes	(2.38)
$S^{(i,s)}(K;c)$, $\hat{S}(K;c)$	suboptimality of one episode and its mean	(2.39)
$D(K;c)$	divergence rate	(2.42)
$\lvert K\rvert$, r	size and rank of a linear policy	Definition 2.6
j	iteration index of a solver	Definition 2.7
g_j, $\bar{g}_j$	convergence signal and its moving mean	Definition 2.7
N_{w}, N_{p}, $\varepsilon_{\mathrm{rel}}$	window, patience and relative tolerance of the convergence rule	Definition 2.7
LQR solvers		
η	step size (learning rate)	Algorithm 1
Σ_K	state correlation matrix of a gain	(3.1)
λ	weight of the nuclear-norm regularizer	(3.3)
K^+, $\bar{r}$	candidate gain of S-PI, number of its singular values	(3.3)
G	gradient mapping	(3.4)
β, ε	step-size reduction factor and stopping tolerance of S-PI	Algorithm 2
σ_{e}, w_k	excitation amplitude and excitation noise for identification	(3.5)
$\hat{n}$, $\hat{C}$	order and output matrix of the identified model	(4.4)
N_{traj}, l, r	number of perturbation pairs, rollout length, smoothing radius	Algorithm 3
Ξ_i	perturbation of the gain	Algorithm 3
$c_k^{\pm}$, $\hat{J}_i^{\pm}$	stage cost and rollout cost of $K \pm \Xi_i$	Algorithm 3
δ_i, $\bar{\delta}$	cost difference of a pair and its mean	Algorithm 3
$\hat{\Sigma}_K$, $\hat{f}(K)$	estimated state correlation and cost	Algorithm 3
Reinforcement learning		
θ, π_θ, μ	actor parameters, policy, mean of the policy	(4.1), (4.3)
ϕ, V_ϕ	critic parameters and critic	Algorithm 6
$r_k(\theta)$, $\hat{A}_k$	probability ratio and advantage estimate	(4.1)
L^{CLIP}, $\varepsilon_{\mathrm{clip}}$	clipped surrogate objective and clip range	(4.1)
γ	discount factor	Section 4.1.2

Symbol	Meaning	Defined in
η_0, N_{env}, $N_{\max}$	initial learning rate, environment steps so far, step cap	(4.2)
W	weight matrix of Linear PPO, $W = -K$	(4.3)
f_ψ, ψ	scalable encoder and its parameters	Algorithm 7
$a_{\max}$	action bound	Algorithm 7
$\mathcal{L}$, $\omega_1, \ldots, \omega_4$	prior loss and its weights	(4.5)
z_k, d_{z}	output of the bottleneck extractor and its dimension	(4.6)
U_{b}, d_{b}	orthonormal bottleneck projection and bottleneck dimension	(4.6)

A.2 Noise Intensities

The settings used for the different noise intensities, obtained by the calibration (2.41) on the non-inflated system. It is important to note that parametric noise is resampled at every step, and that the disturbance noise uses its sinusoidal variant (2.34).

Table A.2: Standard deviation per noise channel and intensity, calibrated to the cost ratio ρ_ℓ of (2.41). Every value reaches its ratio within 0.9 %.

Channel	*none*	*weak*	*standard*	*strong*	*divergent*
Cost ratio ρ_ℓ	1	1.2	1.5	3	10
Measurement σ_{m}	0	0.389	0.607	1.22	2.58
Control σ_{c}	0	1.07	1.67	3.19	5.68
Parametric σ_{p}	0	0.0211	0.0304	0.0451	0.0582
Disturbance σ_{d}	0	0.00767	0.0122	0.0246	0.0519

A.3 Noise Intensities of the Perception Environment

The settings used for the different noise intensities in the perception-based environment of Section 5.2, obtained by the calibration (2.41) with the optimal controller acting on the true state as reference. Measurement noise is not calibrated but set by hand (Section 5.4.1).

Table A.3: Standard deviation per noise channel and intensity, calibrated to the cost ratio ρ_ℓ of (2.41). Every value reaches its ratio within 1.1 %. Measurement noise is set by hand and not calibrated.

Channel	*none*	*weak*	*standard*	*strong*	*divergent*
Cost ratio ρ_ℓ	1	1.2	1.5	3	10
Control σ_c	0	1.31	2.09	4.15	8.78
Parametric σ_p	0	0.0211	0.0286	0.0395	0.0453
Disturbance σ_d	0	0.00202	0.00328	0.00652	0.0138
Measurement σ_m	0	0.0625	0.125	0.25	0.5

Appendix B

Additional Proofs

Appendix C

Complete Simulation Results

Here the results from some hyperparameter tuning are shown. For simplicity, mostly the ones under *large* inflation are shown, unless named otherwise, as this is the most challenging setting from what we have observed. The remaining plots and tables can be found in the repository[CITE: TODO repo link].

C.1 Learning Rate of Model-based PG

Figure C.1 and Table C.1 show the results for *large* inflation.

Table C.1: Model-based PG under *large* inflation, over the seeds.

Learning rate	1e-06	3e-06	1e-05	3e-05	0.0001	0.0012
True cost J	5858 ± 0.001042	5852 ± 0.0003911	5849 ± 0.0001285	5848 ± 0.001456	$1.243 \times 10^4 \pm 1.059$	$1.243 \times 10^4 \pm 1.059$
Training suboptimality $(J - J^*)/J^*$	$0.00173 \pm 1.78 \times 10^{-7}$	$0.000605 \pm 6.69 \times 10^{-8}$	$0.000163 \pm 2.2 \times 10^{-8}$	$4.04 \times 10^{-5} \pm 2.49 \times 10^{-7}$	1.13 ± 0.000181	1.13 ± 0.000181
Test cost	5978 ± 5281	5972 ± 5276	5969 ± 5273	5968 ± 5273	$1.276 \times 10^4 \pm 1.147 \times 10^4$	$1.276 \times 10^4 \pm 1.147 \times 10^4$
Iterations	2390 ± 0	1123 ± 0	454 ± 0	1306 ± 1	64 ± 0	20000 ± 0
Iterations to 1 %	1039 ± 0	342 ± 0	84 ± 0	1040 ± 1	–	–
Env. steps to 1 %	0 ± 0	0 ± 0	0 ± 0	0 ± 0	–	–
Env. steps to convergence	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0

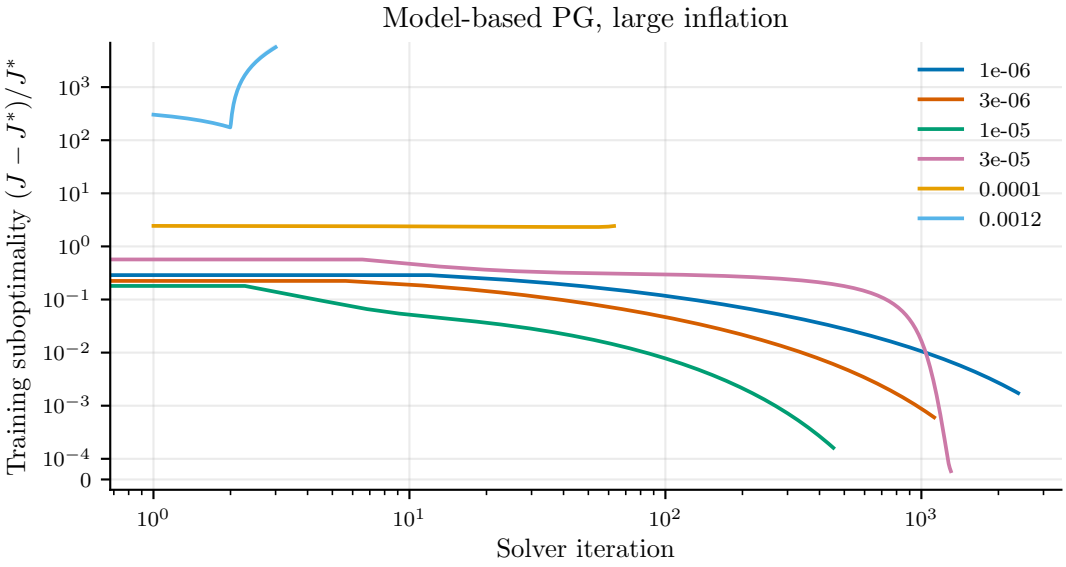


Figure C.1: Training suboptimality of Model-based PG over the solver iterations for each learning rate under *large* inflation. Median and range over ten seeds.

C.2 Nuclear-Norm Weight of S-PI

Table C.2 and Fig. C.2 show the results for *large* inflation.

Table C.2: S-PI under *large* inflation, over the seeds.

λ	1	5	20	50	200	1000
True cost J	5848 ± 0.0009277	$5848 \pm 4.736 \times 10^{-5}$	$5849 \pm 6.53 \times 10^{-5}$	5851 ± 0.0005886	5878 ± 0.0002686	6146 ± 0.0005713
Training suboptimality $(J - J^*)/J^*$	$2.87 \times 10^{-7} \pm 1.59 \times 10^{-7}$	$4.81 \times 10^{-6} \pm 8.1 \times 10^{-9}$	$7.4 \times 10^{-5} \pm 1.12 \times 10^{-8}$	$0.000429 \pm 1.01 \times 10^{-7}$	$0.00506 \pm 4.59 \times 10^{-8}$	$0.0509 \pm 9.77 \times 10^{-8}$
Gain rank	3.8 ± 0.9	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0	1.0 ± 0.0
Test cost	5968 ± 5272	5968 ± 5272	5969 ± 5273	5971 ± 5275	5998 ± 5299	6274 ± 5539
Iterations	155 ± 1	77 ± 1	68 ± 0	72 ± 0	68 ± 0	64 ± 0
Spectral radius	0.9914 ± 0.0000	0.9914 ± 0.0000	0.9916 ± 0.0000	0.9919 ± 0.0000	0.9929 ± 0.0000	0.9952 ± 0.0000

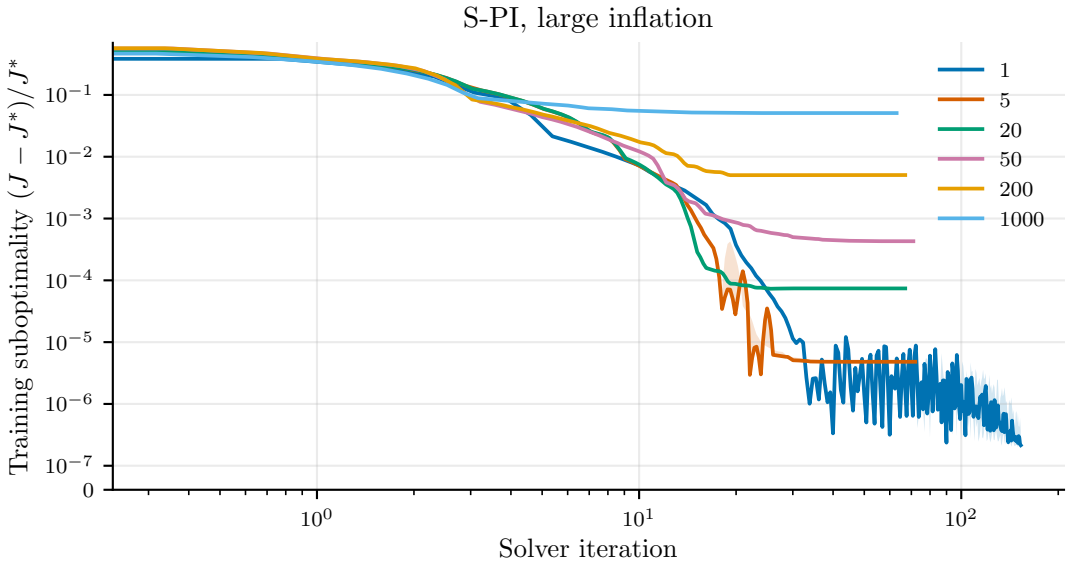


Figure C.2: Training suboptimality of S-PI over the solver iterations for each nuclear-norm weight λ under *large* inflation. Median and range over ten seeds.

C.3 N4SID Identification Length

Figures C.3 and C.4 show the results for *large* and *medium* inflation over all trajectory lengths and Hankel horizons, Tables C.3 and C.4 the detailed results for a Hankel horizon of five.

Table C.3: N4SID + Model-based PG on a system with *medium* inflation with a Hankel horizon of 5.

Trajectory length T	500	1000	2000	4000	8000
Training suboptimality $(J - J^*)/J^*$	$3.15 \times 10^{-5} \pm 2.09 \times 10^{-7}$	$3.17 \times 10^{-5} \pm 2.48 \times 10^{-7}$	$3.16 \times 10^{-5} \pm 2.06 \times 10^{-7}$	$3.15 \times 10^{-5} \pm 3.6 \times 10^{-7}$	$3.16 \times 10^{-5} \pm 1.96 \times 10^{-7}$
Iterations	346 ± 1	346 ± 1	346 ± 1	346 ± 1	346 ± 1
Env. steps to convergence	500 ± 0	$10^3 \pm 0$	$2 \times 10^3 \pm 0$	$4 \times 10^3 \pm 0$	$8 \times 10^3 \pm 0$
Env. steps to best policy	500 ± 0	$10^3 \pm 0$	$2 \times 10^3 \pm 0$	$4 \times 10^3 \pm 0$	$8 \times 10^3 \pm 0$
Wall time (s)	1.1 ± 0.1	1.2 ± 0.0	1.4 ± 0.0	1.9 ± 0.0	2.9 ± 0.1

Table C.4: N4SID + Model-based PG on a system with *large* inflation with a Hankel horizon of 5. Identified counts the seeds a model was obtained from and stable those whose controller left the plant stable.

Trajectory length T	500	1000	2000	4000	8000
Identified	0/10	10/10	10/10	10/10	10/10
Stable	0/10	10/10	10/10	10/10	10/10
Training suboptimality $(J - J^*)/J^*$	–	0.00382 ± 0.00269	$3.21 \times 10^{-5} \pm 4.14 \times 10^{-7}$	$3.18 \times 10^{-5} \pm 1.12 \times 10^{-7}$	$3.18 \times 10^{-5} \pm 8.71 \times 10^{-8}$
Iterations	–	347 ± 9	346 ± 0	346 ± 0	346 ± 0
Env. steps to convergence	–	$10^3 \pm 0$	$2 \times 10^3 \pm 0$	$4 \times 10^3 \pm 0$	$8 \times 10^3 \pm 0$
Env. steps to best policy	–	$10^3 \pm 0$	$2 \times 10^3 \pm 0$	$4 \times 10^3 \pm 0$	$8 \times 10^3 \pm 0$
Wall time (s)	–	23.0 ± 1.0	26.4 ± 0.6	29.7 ± 0.9	36.4 ± 2.0

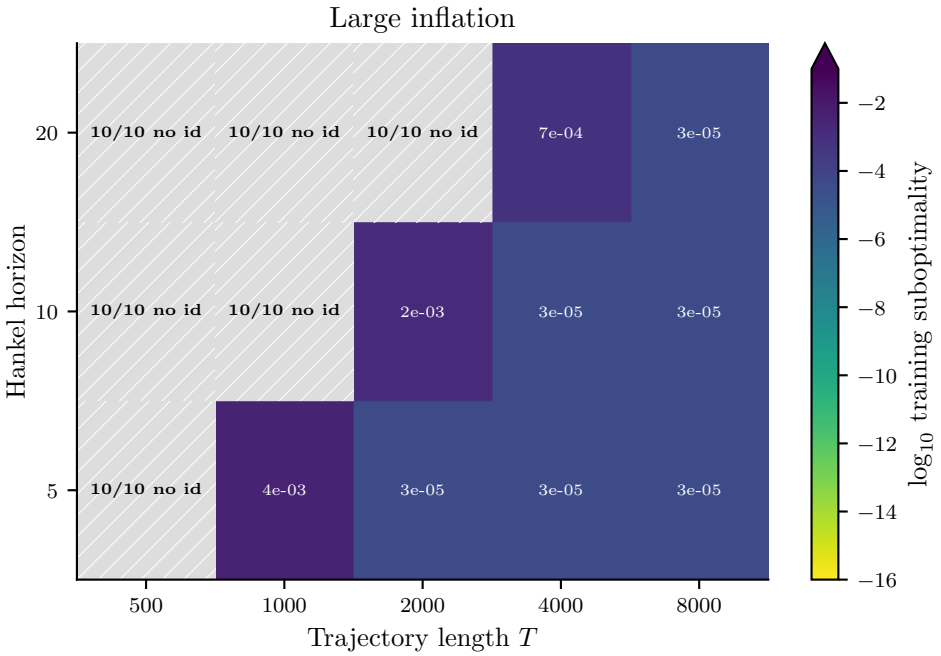


Figure C.3: Median training suboptimality of N4SID + Model-based PG per trajectory length T and Hankel horizon under *large* inflation.

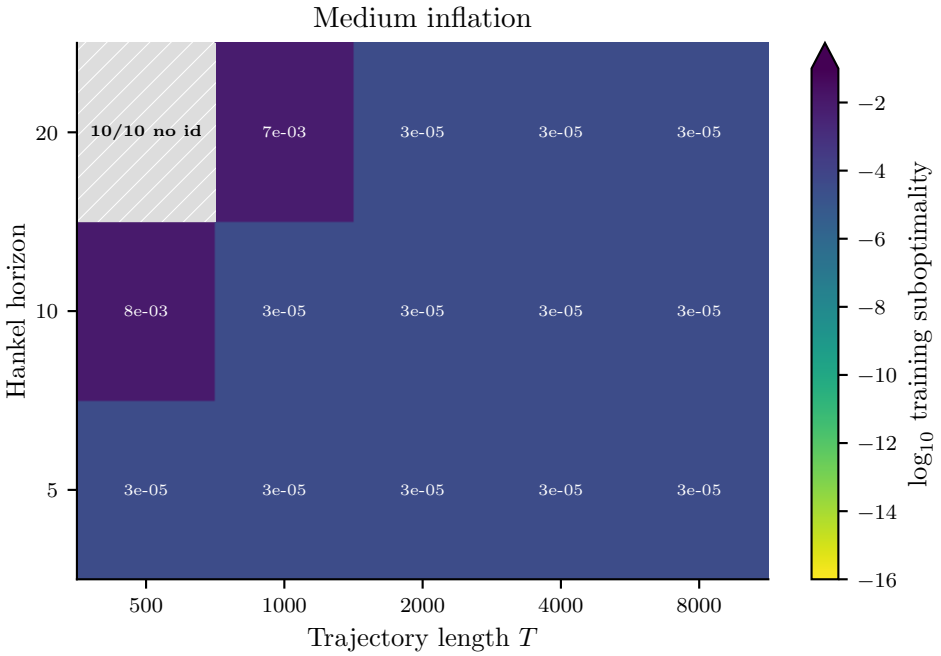


Figure C.4: Median training suboptimality of N4SID + Model-based PG per trajectory length T and Hankel horizon under *medium* inflation.

C.4 Sample Size

Figures C.5 and C.6 show the results of Model-free NPG for *large* inflation, Table C.5 the training suboptimality for all inflation sizes.

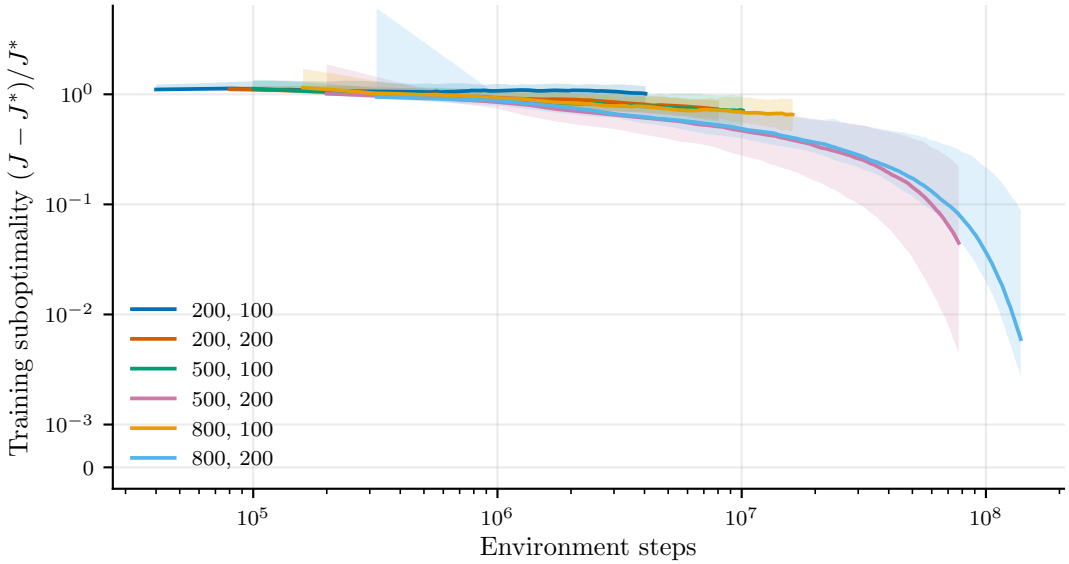


Figure C.5: Training suboptimality of Model-free NPG over the environment steps for each rollout length l and number of trajectories N_{traj} under *large* inflation. Median and range over ten seeds.

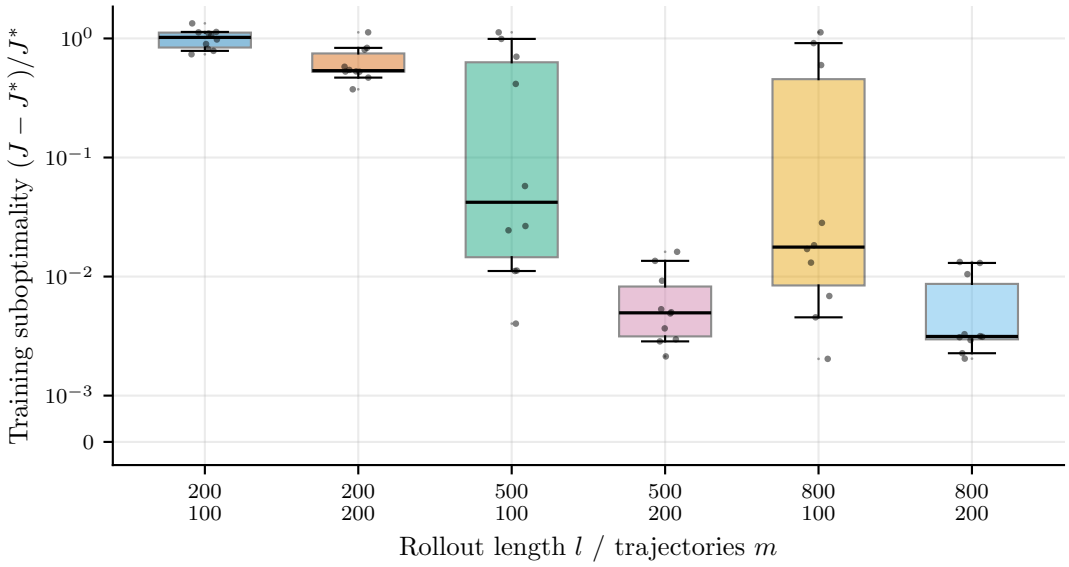


Figure C.6: Final training suboptimality of Model-free NPG for each rollout length l and number of trajectories N_{traj} under *large* inflation.

Table C.5: Model-free NPG: training suboptimality $(J - J^*)/J^*$ per inflation size (rows) and sample-size combination (columns).

(l, m)	200, 100	200, 200	500, 100	500, 200	800, 100	800, 200
base	0.1868 ± 0.329	0.078 ± 0.003253	0.00253 ± 0.001248	0.003342 ± 0.0009051	0.002967 ± 0.001213	0.003835 ± 0.00146
small	0.3514 ± 0.2896	0.2185 ± 0.06357	0.003464 ± 0.002515	0.002264 ± 0.002548	0.00296 ± 0.002492	0.00183 ± 0.001281
medium	0.4981 ± 0.2409	0.2816 ± 0.0991	0.007411 ± 0.007269	0.002158 ± 0.001353	0.1171 ± 0.3566	0.2279 ± 0.4766
large	0.9981 ± 0.1889	0.631 ± 0.2237	0.3369 ± 0.4452	0.006577 ± 0.004818	0.2726 ± 0.4367	0.005664 ± 0.004618

C.5 Learning Rate of Model-free NPG

Figure C.7 and Table C.6 show the results for *large* inflation.

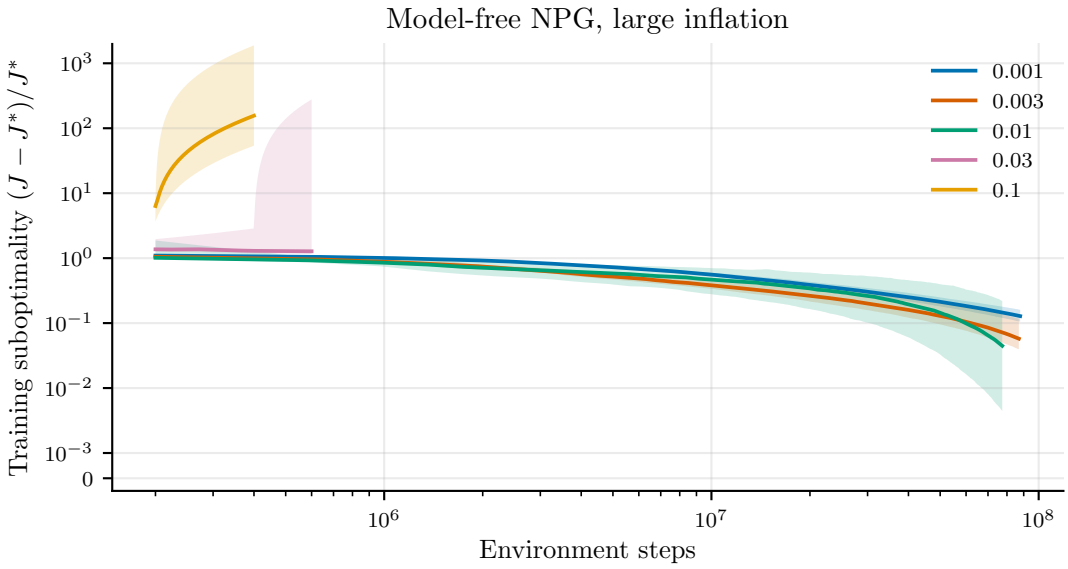


Figure C.7: Training suboptimality of Model-free NPG over the environment steps for each learning rate under *large* inflation. Median and range over ten seeds.

Table C.6: Model-free NPG under *large* inflation, over the seeds.

Learning rate	0.001	0.003	0.01	0.03	0.1
True cost J	6423 ± 150.4	6074 ± 100.1	5887 ± 28.18	9906 ± 3502	$1.243 \times 10^4 \pm 1.059$
Training suboptimality $(J - J^*)/J^*$	0.0982 ± 0.0257	0.0386 ± 0.0171	0.00658 ± 0.00482	0.694 ± 0.599	1.13 ± 0.000181
Test cost	6555 ± 5793	6198 ± 5476	6007 ± 5306	$1.032 \times 10^4 \pm 1.04 \times 10^4$	$1.276 \times 10^4 \pm 1.147 \times 10^4$
Iterations	617 ± 112	617 ± 82	610 ± 104	390 ± 373	101 ± 0
Iterations to 1 %	–	–	509 ± 110	700 ± 36	–
Env. steps to 1 %	–	–	$1.02 \times 10^8 \pm 2.2 \times 10^7$	$1.4 \times 10^8 \pm 7.29 \times 10^6$	–
Env. steps to convergence	$1.23 \times 10^8 \pm 2.23 \times 10^7$	$1.23 \times 10^8 \pm 1.64 \times 10^7$	$1.22 \times 10^8 \pm 2.08 \times 10^7$	$7.41 \times 10^7 \pm 7.83 \times 10^7$	$6.87 \times 10^5 \pm 7.44 \times 10^4$

C.6 Step Size and Nuclear-Norm Weight of S-PI (Model-Free)

Here, we show different combinations of values for the stepsize and λ under *large* inflation. Excluded are the tables for $\lambda = 400$, as no rank reduction could be

observed (λ too low), as well as $\lambda = 4 \times 10^6$, as here all produced controllers were unstable (λ too high).

From the remainders, we picked a value that exhibits rank reduction, while at the same time having a low suboptimality. Tables C.7 to C.9 show the results for the remaining values of λ.

Table C.7: S-PI (model-free) with $\lambda = 4000$ under *large* inflation, over the seeds.

Step size	3.5e-10	3.5e-09	3.5e-08	3.5e-07	3.5e-06
True cost J	$1.241 \times 10^4 \pm 18.05$	$1.213 \times 10^4 \pm 245.3$	9581 ± 30.62	8030 ± 19.81	8956 ± 103.1
Training suboptimality $(J - J^*)/J^*$	1.12 ± 0.00309	1.07 ± 0.042	0.638 ± 0.00524	0.373 ± 0.00339	0.531 ± 0.0176
Gain rank	100.0 ± 0.0	100.0 ± 0.0	100.0 ± 0.0	96.7 ± 2.1	81.7 ± 5.0
Spectral radius	0.9975 ± 0.0000	0.9975 ± 0.0000	0.9976 ± 0.0000	0.9979 ± 0.0000	0.9983 ± 0.0000
Iterations	50 ± 0	50 ± 0	50 ± 0	50 ± 0	50 ± 0
Env. steps to convergence	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$
Env. steps to best policy	$2.8 \times 10^6 \pm 3.11 \times 10^6$	$5.7 \times 10^6 \pm 4.6 \times 10^6$	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$

Table C.8: S-PI (model-free) with $\lambda = 40000$ under *large* inflation, over the seeds.

Step size	3.5e-10	3.5e-09	3.5e-08	3.5e-07	3.5e-06
True cost J	$1.238 \times 10^4 \pm 16.65$	$1.195 \times 10^4 \pm 11.76$	9927 ± 33	$7.525 \times 10^4 \pm 1.206 \times 10^4$	$4.124 \times 10^4 \pm 1.244 \times 10^4$
Training suboptimality $(J - J^*)/J^*$	1.12 ± 0.00285	1.04 ± 0.00201	0.698 ± 0.00564	11.9 ± 2.06	6.05 ± 2.13
Gain rank	100.0 ± 0.0	100.0 ± 0.0	53.3 ± 4.5	58.8 ± 4.9	28.1 ± 10.7
Spectral radius	0.9975 ± 0.0000	0.9975 ± 0.0000	0.9979 ± 0.0000	0.9999 ± 0.0000	0.9997 ± 0.0001
Iterations	50 ± 0	50 ± 0	50 ± 0	50 ± 0	50 ± 0
Env. steps to convergence	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$	$10^7 \pm 0$
Env. steps to best policy	$9.02 \times 10^6 \pm 3.1 \times 10^6$	$10^7 \pm 0$	$10^7 \pm 0$	$9.64 \times 10^6 \pm 3.1 \times 10^5$	$1.22 \times 10^6 \pm 1.14 \times 10^5$

Table C.9: S-PI (model-free) with $\lambda = 400000$ under *large* inflation, over the seeds. Stable counts the seeds whose controller left the plant stable.

Step size	3.5e-10	3.5e-09	3.5e-08	3.5e-07	3.5e-06
Stable	10/10	10/10	0/10	0/10	0/10
True cost J	$1.242 \times 10^4 \pm 4.022$	$1.245 \times 10^4 \pm 28.56$	–	–	–
Training suboptimality $(J - J^*)/J^*$	1.12 ± 0.000688	1.13 ± 0.00488	–	–	–
Gain rank	100.0 ± 0.0	15.6 ± 0.5	–	–	–
Spectral radius	0.9975 ± 0.0000	0.9978 ± 0.0000	–	–	–
Iterations	50 ± 0	50 ± 0	–	–	–
Env. steps to convergence	$10^7 \pm 0$	$10^7 \pm 0$	–	–	–
Env. steps to best policy	$10^7 \pm 0$	$10^7 \pm 0$	–	–	–

C.7 Noise on the Identified Model

Table C.10 and Fig. C.8 show the results of model-free approaches under *large* inflation and *strong* noise.

Table C.10: Suboptimality S under *large* inflation and each noise channel at intensity *strong*.

Channel	Baseline	N4SID + DARE	N4SID + Model-based PG	N4SID + S-PI	Model-free NPG	S-PI (model-free)
none	+0	$+1.54 \times 10^{-7} \pm 2.76 \times 10^{-7}$	$+3.91 \times 10^{-5} \pm 2.43 \times 10^{-5}$	$+8.53 \times 10^{-6} \pm 5.31 \times 10^{-6}$	$+0.00755 \pm 0.00667$	$+5.99 \pm 7.72$
measurement	+0	$+7.22 \times 10^{-6} \pm 4.73 \times 10^{-5}$	$+0.0238 \pm 0.00659$	-0.00955 ± 0.00266	$+0.209 \pm 0.0943$	$+0.259 \pm 0.777$
control	+0	$+7.51 \times 10^{-6} \pm 5.97 \times 10^{-5}$	-0.00548 ± 0.0023	-0.00292 ± 0.00105	-0.0246 ± 0.0361	$+1.58 \pm 2.16$
parametric	+0	$+4.37 \times 10^{-7} \pm 5.97 \times 10^{-5}$	-0.000895 ± 0.0102	-0.00046 ± 0.00481	$+0.0289 \pm 0.181$	$+7.18 \times 10^{4} \pm 1.25 \times 10^{6}$
disturbance	+0	$+1.44 \times 10^{-7} \pm 4.18 \times 10^{-5}$	$+0.0047 \pm 0.00131$	$+0.00205 \pm 0.000596$	-0.0524 ± 0.0263	$+0.927 \pm 0.859$
all channels	+0	$+1.87 \times 10^{-5} \pm 0.000192$	$+0.000241 \pm 0.0421$	-0.0112 ± 0.0152	$+0.209 \pm 0.18$	$+535 \pm 6.25 \times 10^{3}$

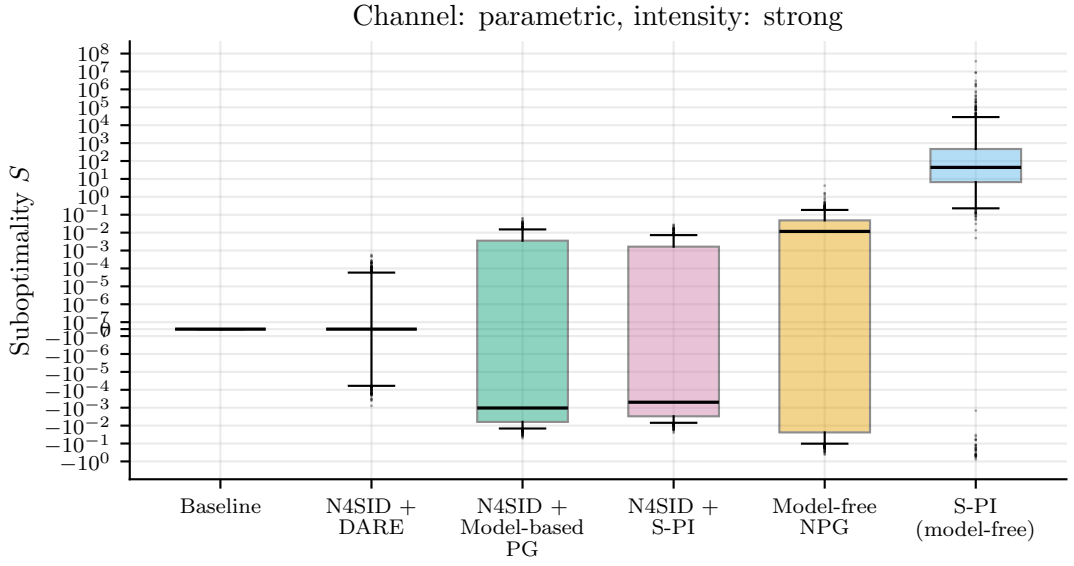


Figure C.8: TODO: Suboptimality S of the model-free solvers under *large* inflation and parametric uncertainty at intensity *strong*.

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